A General Framework for Low Soundness Homomorphism Testing

ACCESS



TUSHANT MITTAL
STANFORD UNIVERSITY



SOURYA ROY
UNIVERSITY OF IOWA

Today’s Agenda

Today's Agenda

Introduction

Today's Agenda

Introduction ▶ Homomorphism Testing

Today's Agenda

Introduction ▶ Homomorphism Testing ▶ General Framework

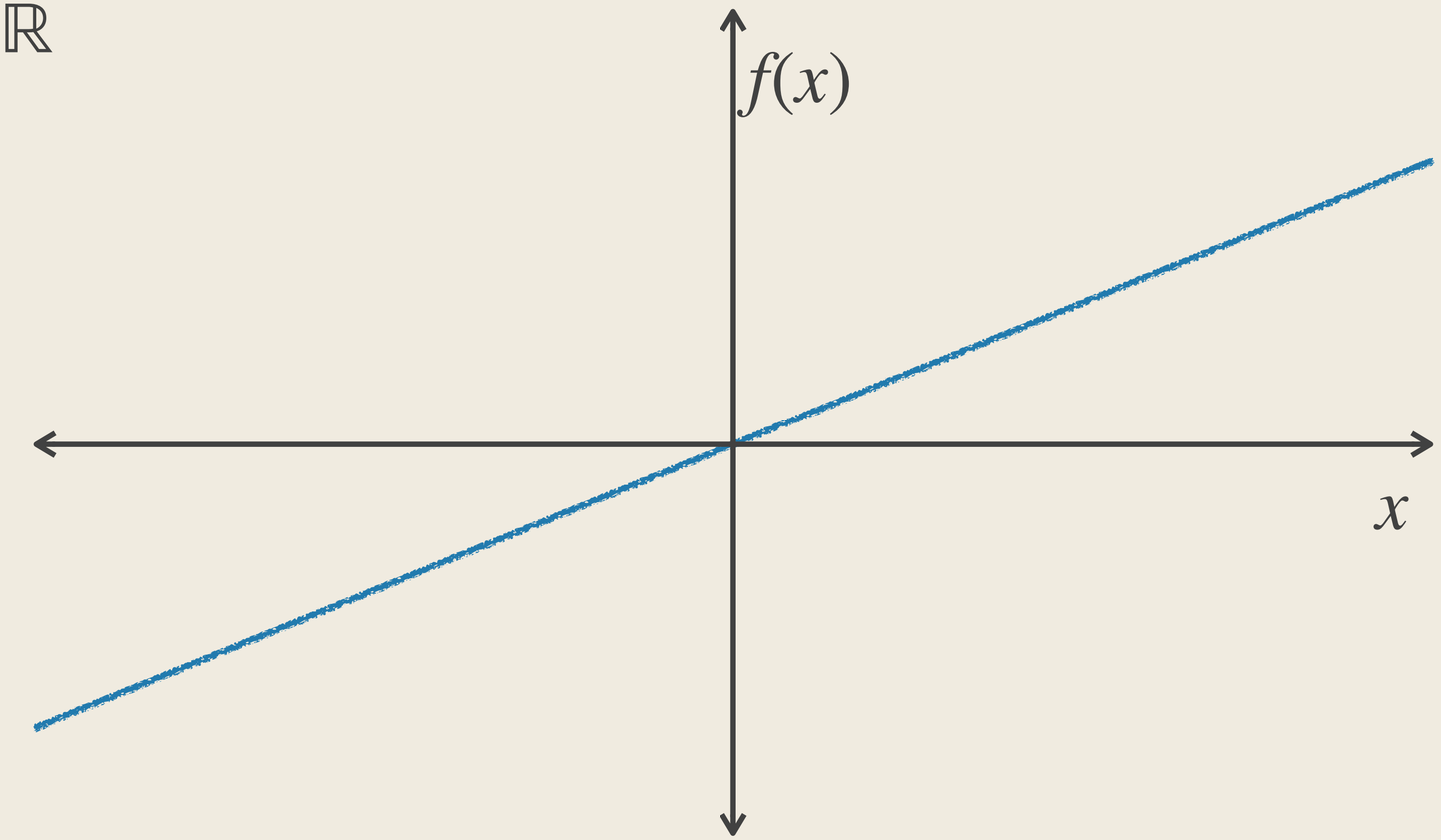
Today's Agenda

Introduction ▶ Homomorphism Testing ▶ General Framework ▶ Our Results

What is a linear function?

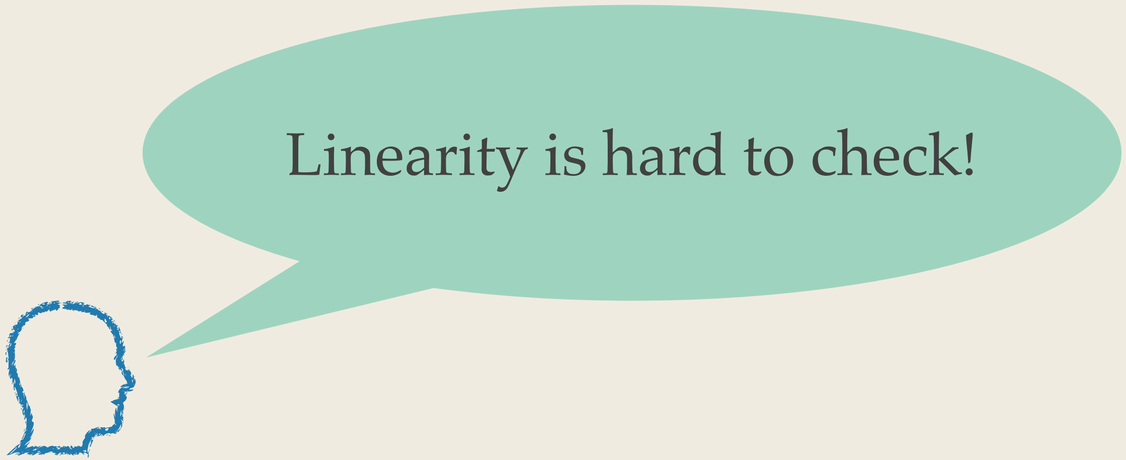
What is a linear function?

$f : \mathbb{R} \to \mathbb{R}$

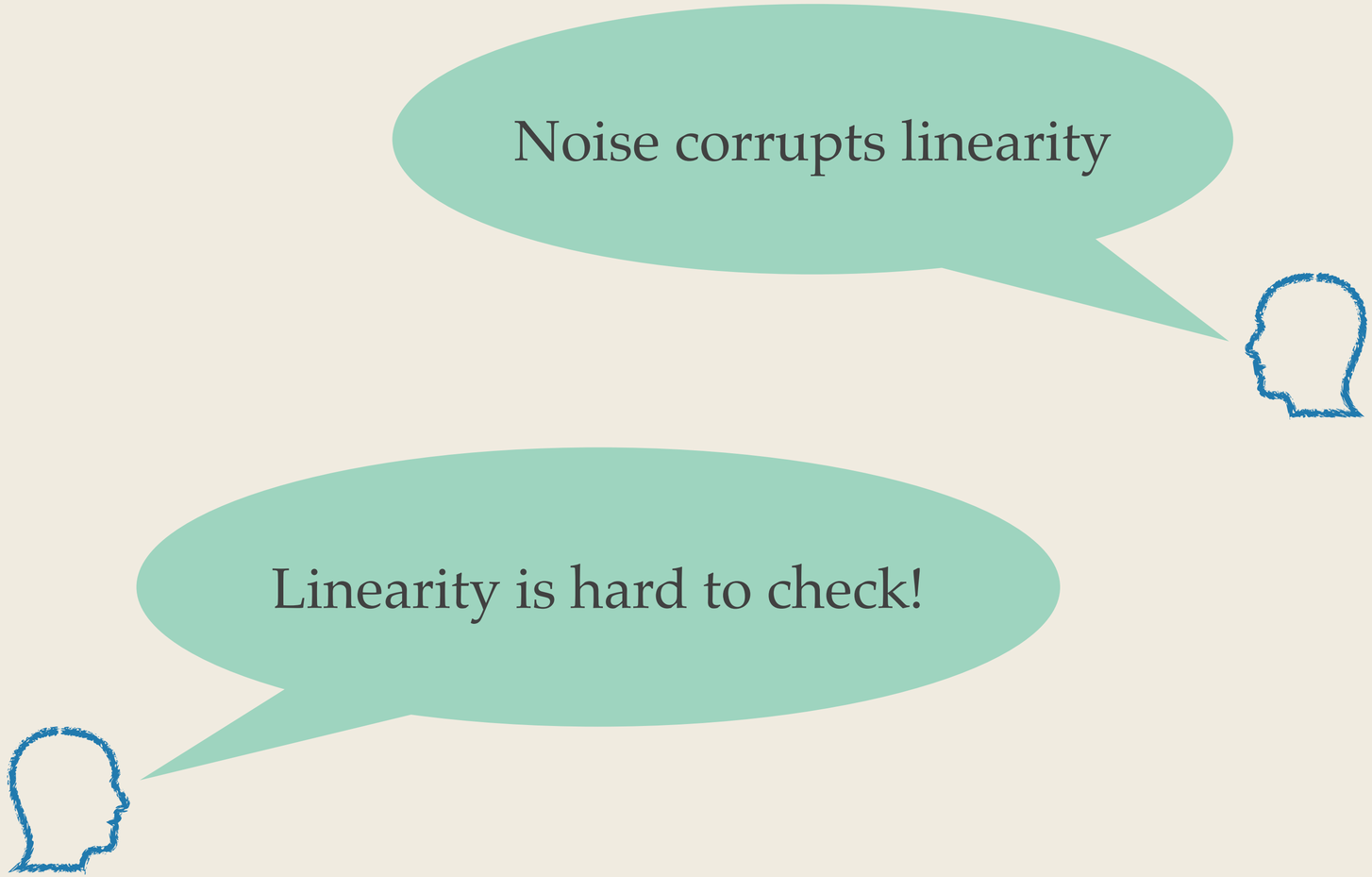


An *approximately* linear function?

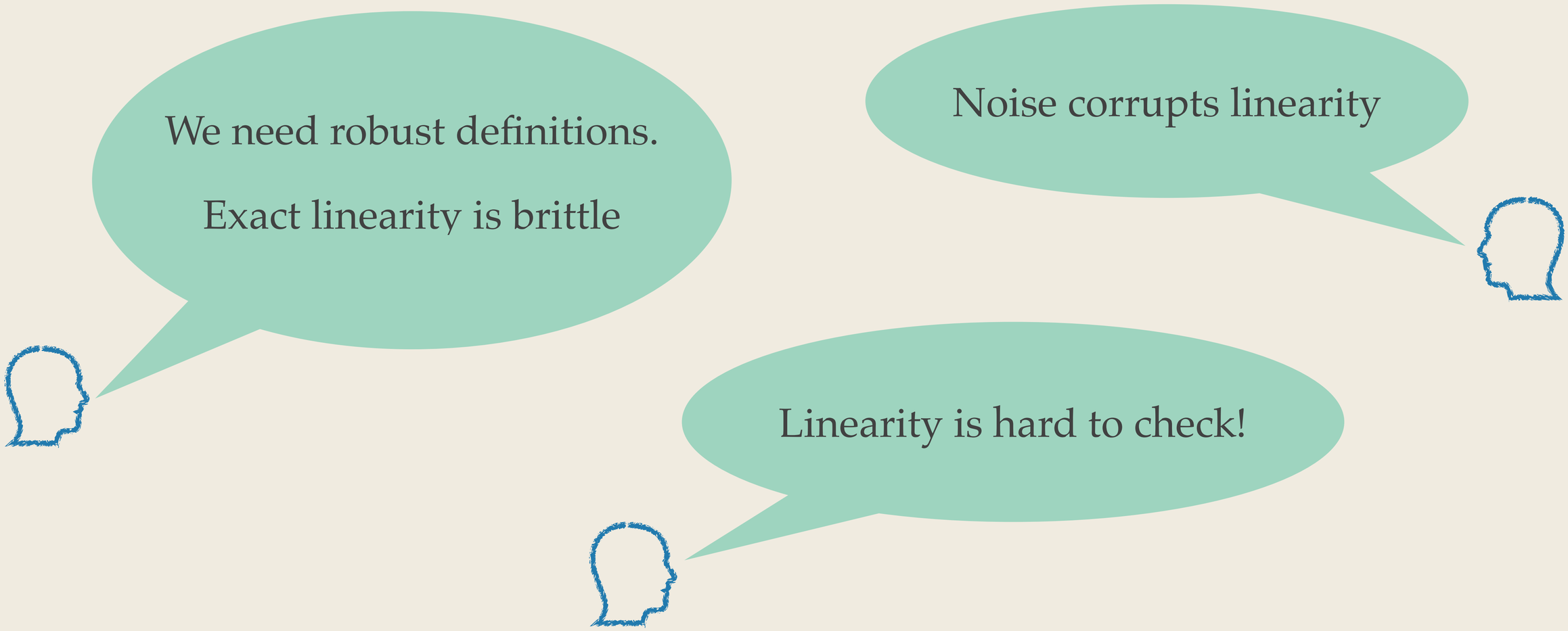
An *approximately* linear function?



An *approximately* linear function?



An *approximately* linear function?



A CS notion of *approximately* linear

A CS notion of *approximately* linear

Algorithmically Efficient

Checkable in constant queries

A CS notion of *approximately* linear

Algorithmically Efficient

Checkable in constant queries

Mathematically Useful

f is close to a true linear function

Motivating Application: PCPs

Motivating Application: PCPs

- Interactive proofs:

Motivating Application: PCPs

- Interactive proofs:
  - Classical – *Approximate Linearity* aka Linearity Tests.

Motivating Application: PCPs

- Interactive proofs:
  - Classical – *Approximate Linearity* aka Linearity Tests.
  - Quantum – *Approximate Homomorphisms* aka Homomorphism Tests.

“Probabilistically

Checkable

Proof (PCP)”



Kristina Armitage/ Quanta Magazine

Probabilistically Checkable Proof (PCP)

Probabilistically Checkable Proof (PCP)

- A PCP verifier is an algorithm that reads $3k$ random bits of the proof.

Probabilistically Checkable Proof (PCP)

- A PCP verifier is an algorithm that reads $3k$ random bits of the proof.
- Outputs Yes/No.

Probabilistically Checkable Proof (PCP)

- A PCP verifier is an algorithm that reads $3k$ random bits of the proof.
- Outputs Yes/No.
  - (Completeness) If the proof is correct, it always outputs YES.

Probabilistically Checkable Proof (PCP)

- A PCP verifier is an algorithm that reads $3k$ random bits of the proof.
- Outputs Yes/No.
  - (Completeness) If the proof is correct, it always outputs YES.
  - (Soundness) If the proof is wrong, it outputs YES with probability $< 2^{-k}$.

Probabilistically Checkable Proof (PCP)

- A PCP verifier is an algorithm that reads $3k$ random bits of the proof.
- Outputs Yes/No.
  - (Completeness) If the proof is correct, it always outputs YES.
  - (Soundness) If the proof is wrong, it outputs YES with probability $< 2^{-k}$.

Probabilistically Checkable Proof (PCP)

- A PCP verifier is an algorithm that reads $3k$ random bits of the proof.
- Outputs Yes/No.
  - (Completeness) If the proof is correct, it always outputs YES.
  - (Soundness) If the proof is wrong, it outputs YES with probability $< 2^{-k}$.

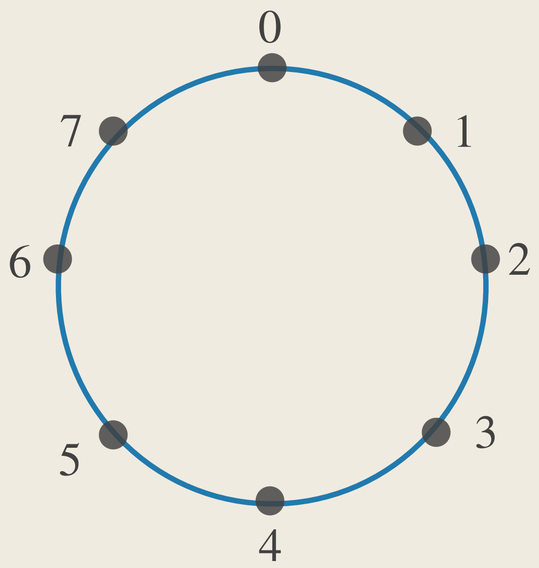
- [Arora-Safra, Arora-Lund-Motwani-Sudan-Szegedy'91]
  Every NP-proof can be converted to a (polynomially large) PCP proof.

Homomorphsim Testing

Groups

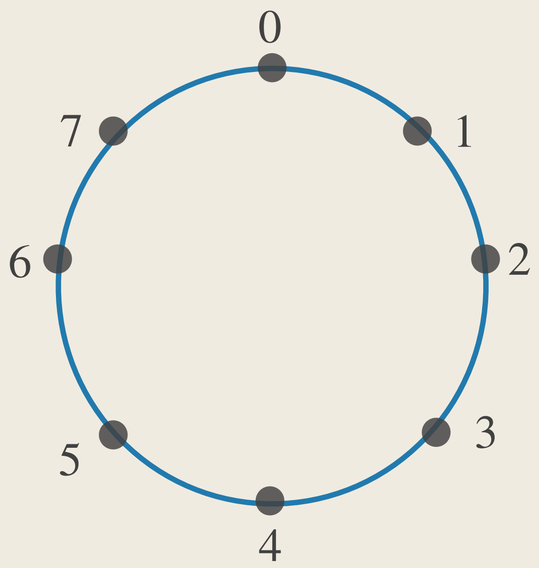
Group: Set & * Operation

Groups



Group: Set & * Operation

Groups



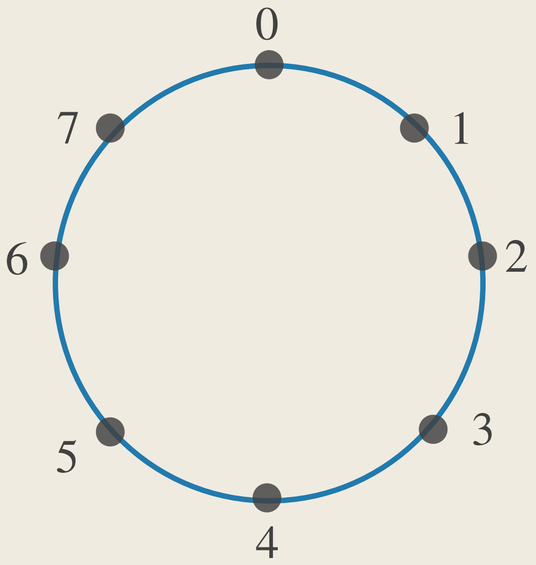
Cyclic Group ($\mathbb{Z}_n$)

Addition modulo n

$$1 * 7 = 0 \mod 8$$

Group: Set & * Operation

Groups

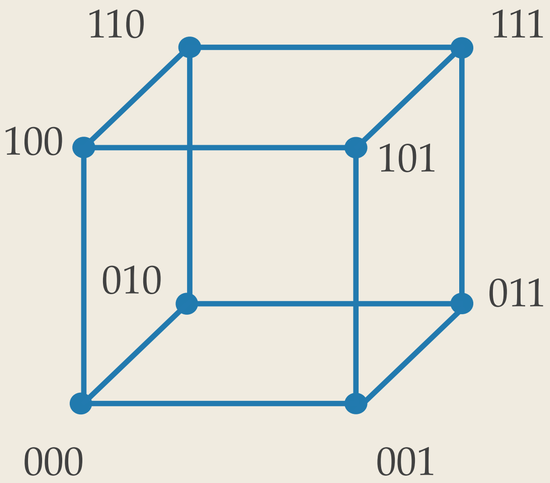


Cyclic Group ($\mathbb{Z}_n$)

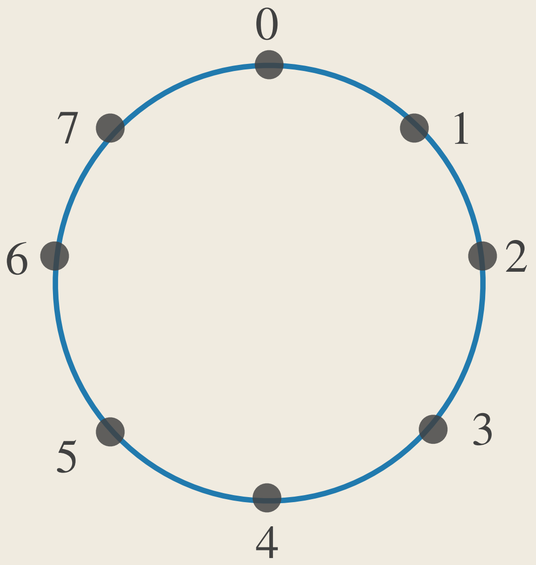
Addition modulo n

$$1 * 7 = 0 \mod 8$$

Group: Set & * Operation



Groups

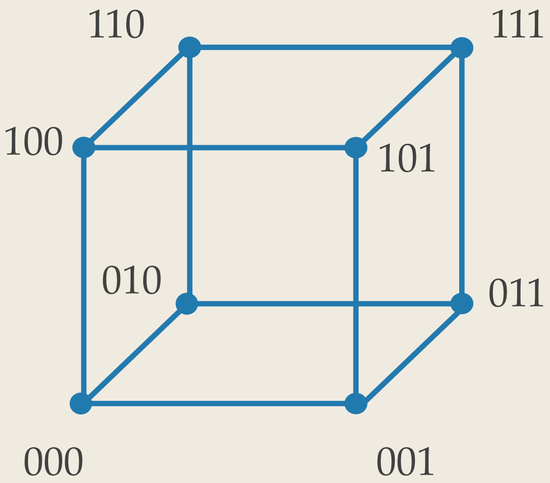


Cyclic Group ($\mathbb{Z}_n$)

Addition modulo n

$1 * 7 = 0 \mod 8$

Group: Set & * Operation



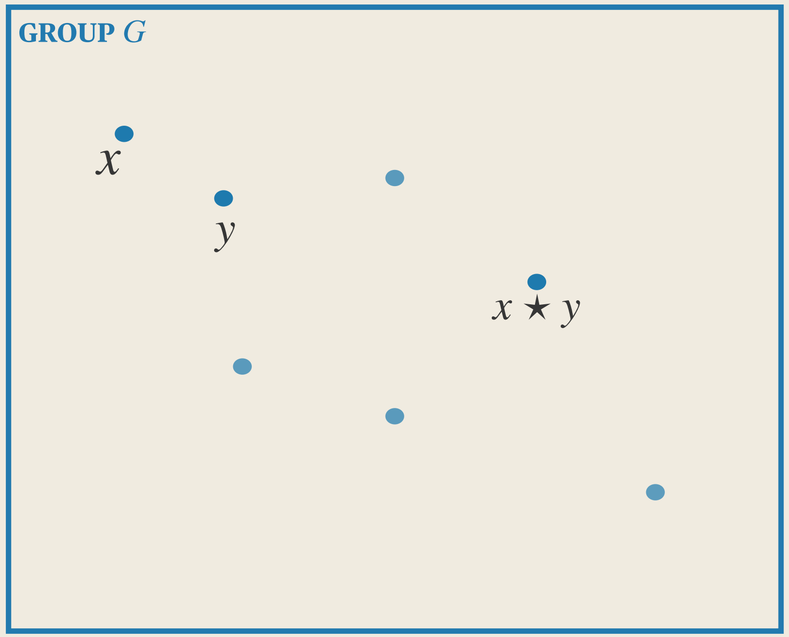
Hypercube ($\mathbb{Z}_2^n$)

Addition mod 2

$001 * 111 = 110.$

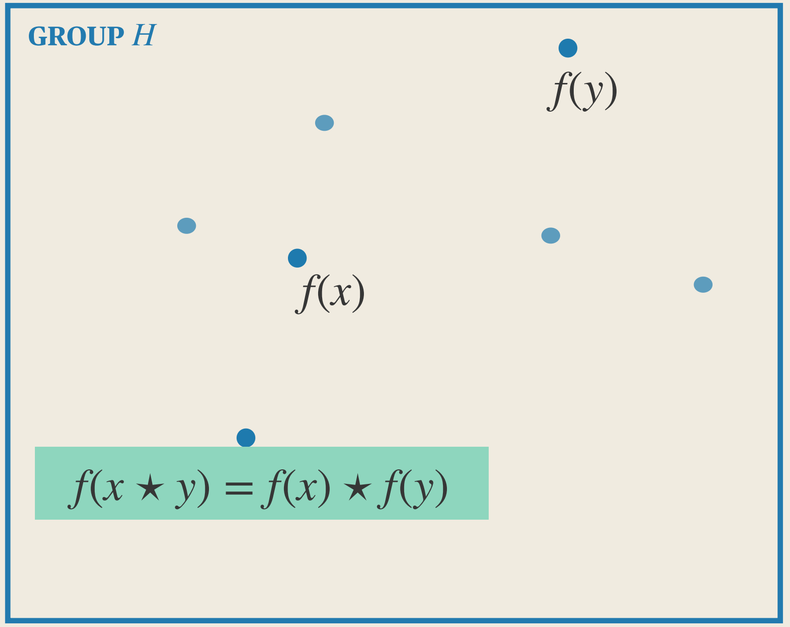
Group Homomorphisms

Group Homomorphisms

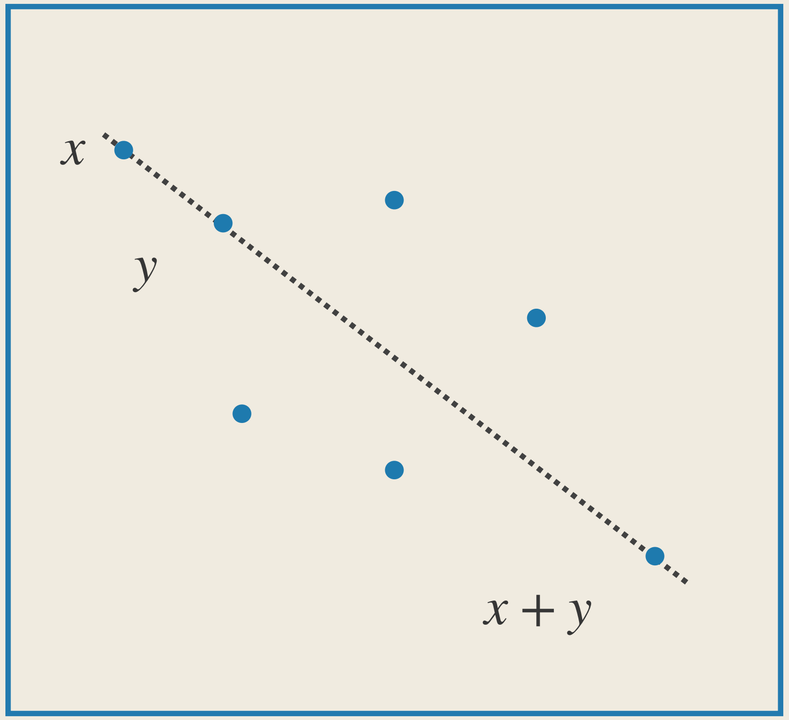


f : Homomorphism

"Preserves group operation"

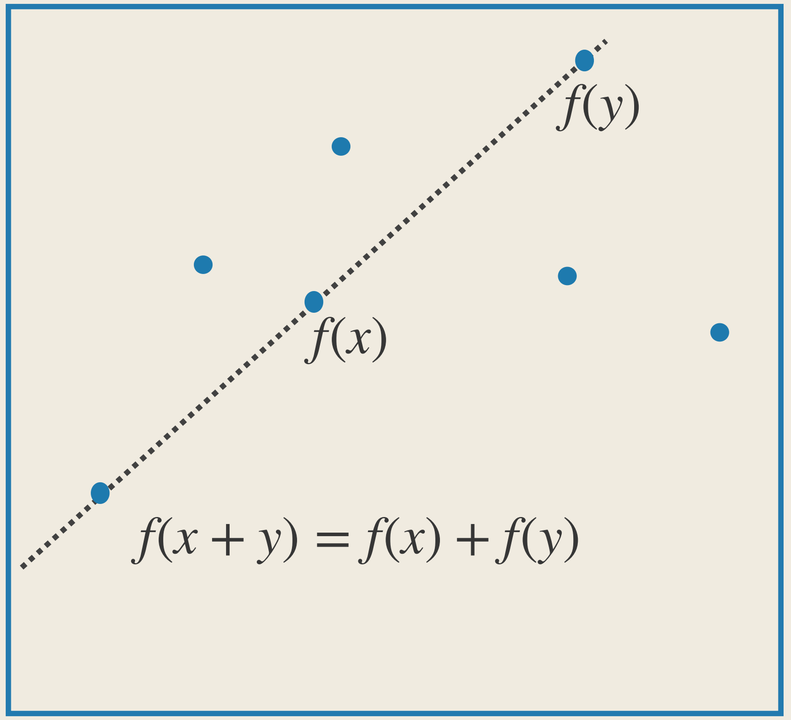


Eg: Linear functions



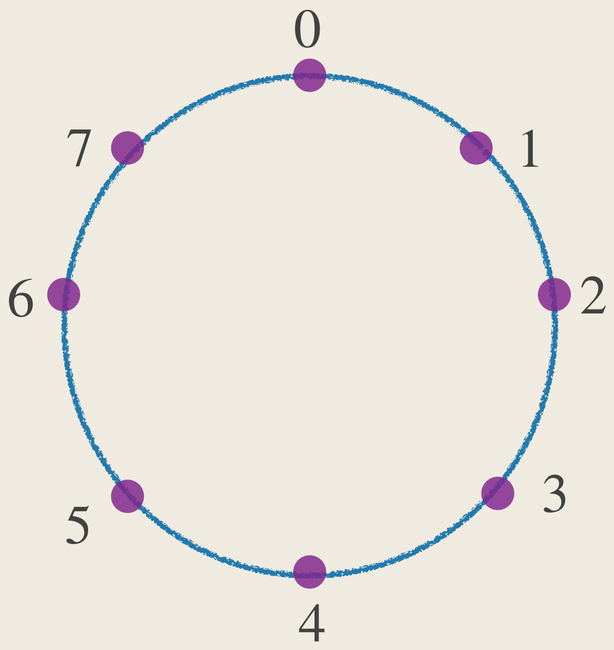
f : a linear function

"Preserves linearity"

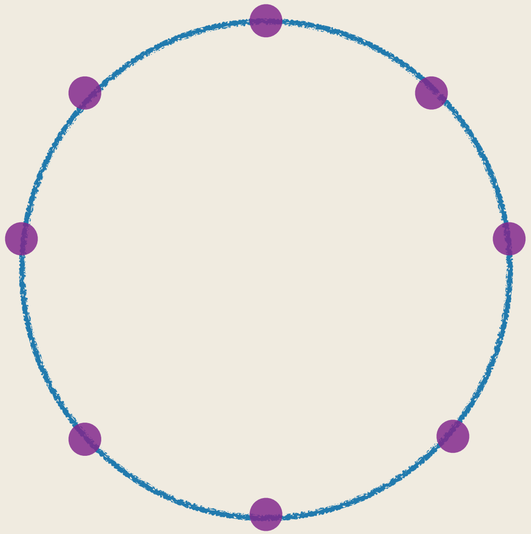


Group Homomorphism: An Example

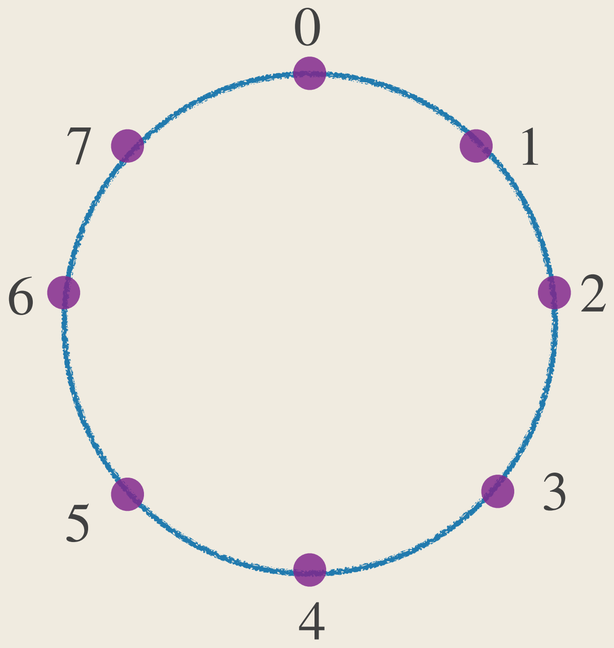
Group Homomorphism: An Example



$$f : \mathbb{Z}_8 \to \mathbb{Z}_8$$

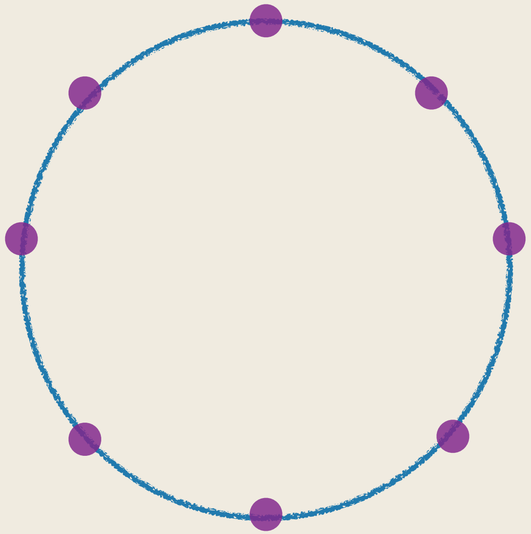


Group Homomorphism: An Example

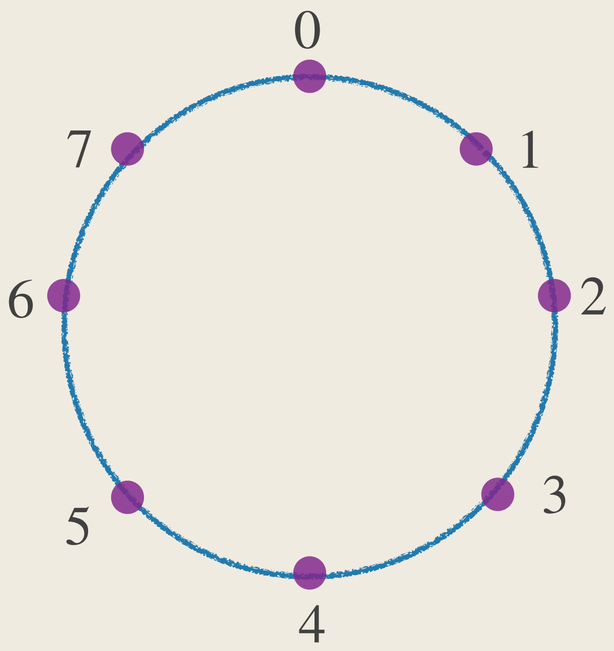


$$f : \mathbb{Z}_8 \to \mathbb{Z}_8$$

$$f(i) = f(1 + \ldots + 1) = i \cdot f(1)$$



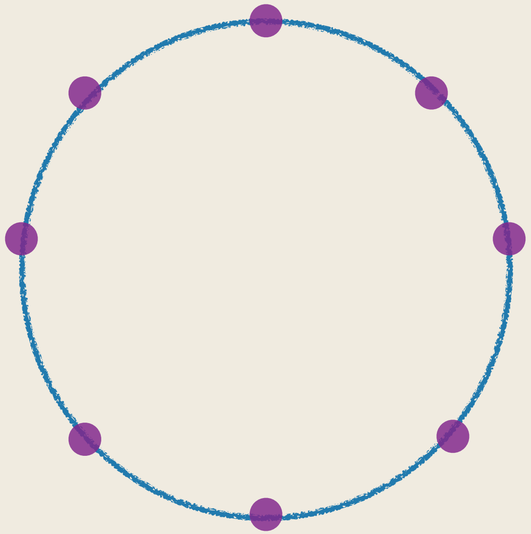
Group Homomorphism: An Example



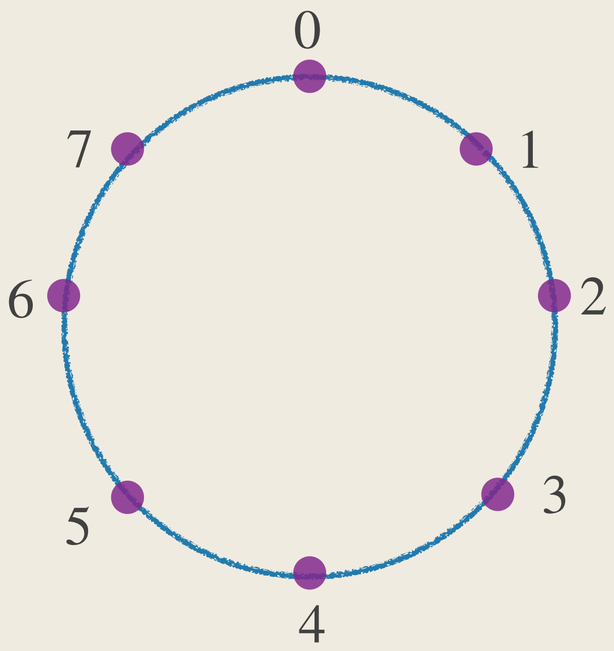
$$f : \mathbb{Z}_8 \to \mathbb{Z}_8$$

$$f(i) = f(1 + \ldots + 1) = i \cdot f(1)$$

$f(0) = 0$

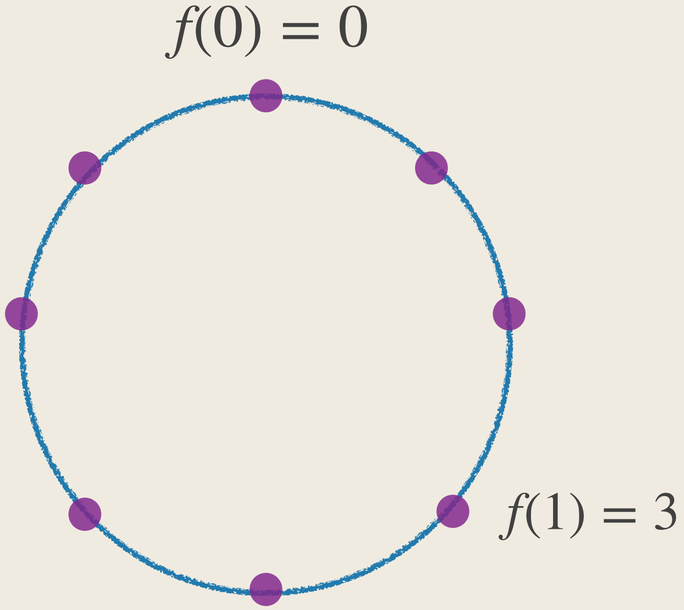


Group Homomorphism: An Example

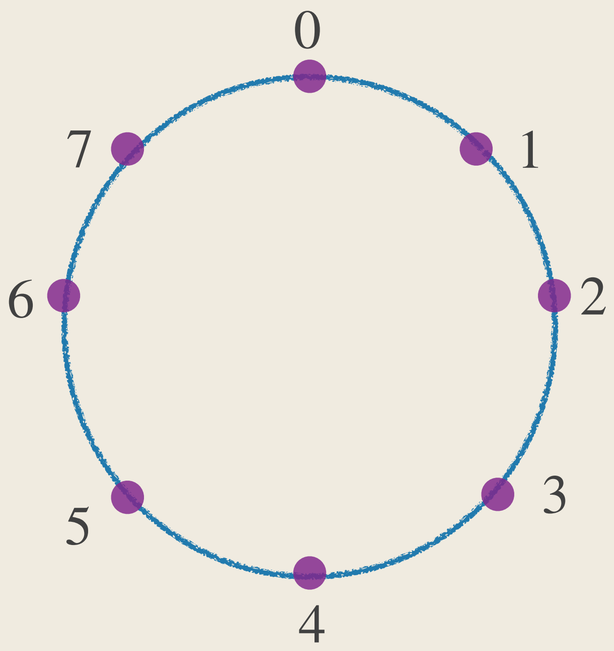


$$f : \mathbb{Z}_8 \to \mathbb{Z}_8$$

$$f(i) = f(1 + \ldots + 1) = i \cdot f(1)$$

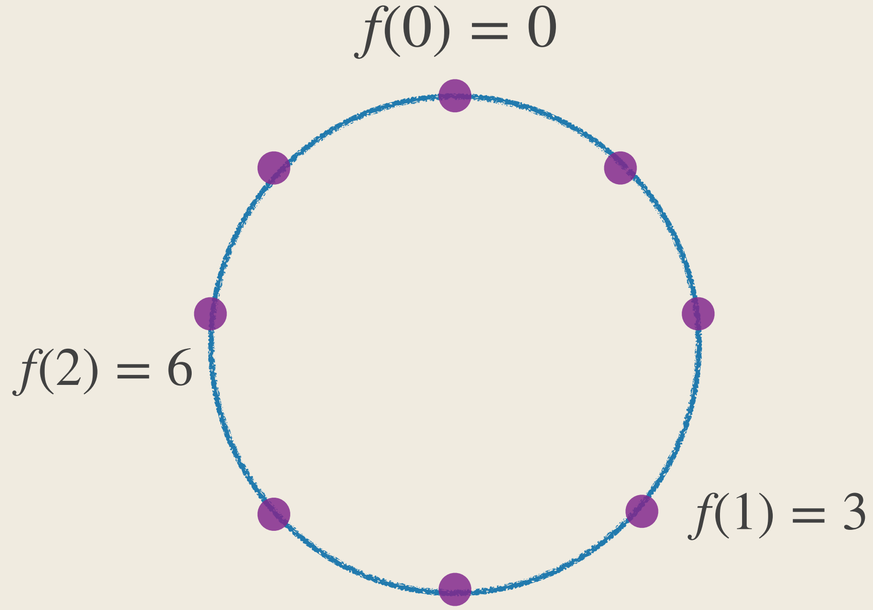


Group Homomorphism: An Example

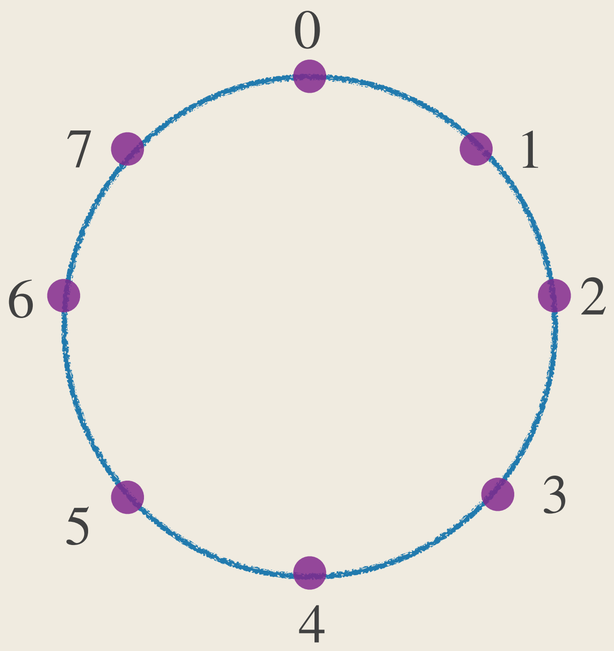


$$f : \mathbb{Z}_8 \to \mathbb{Z}_8$$

$$f(i) = f(1 + \ldots + 1) = i \cdot f(1)$$

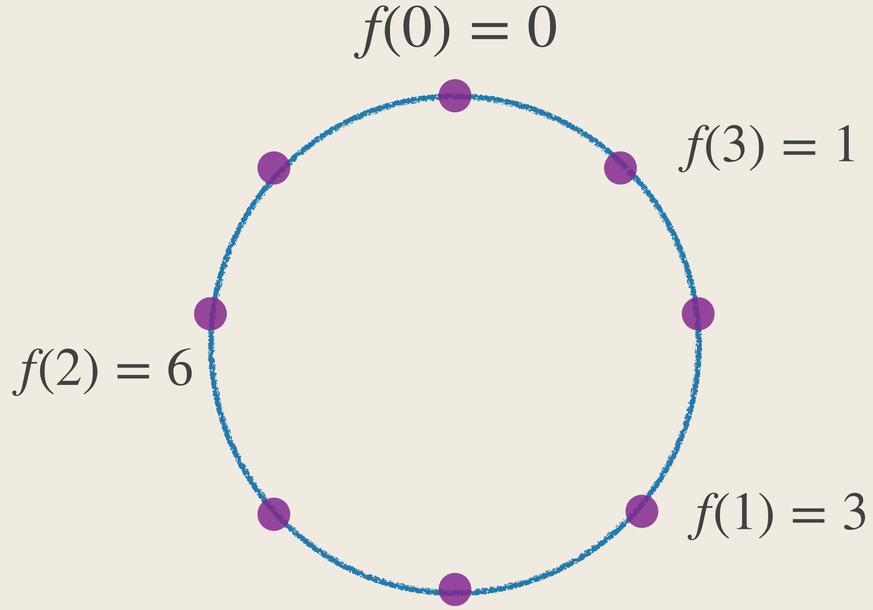


Group Homomorphism: An Example

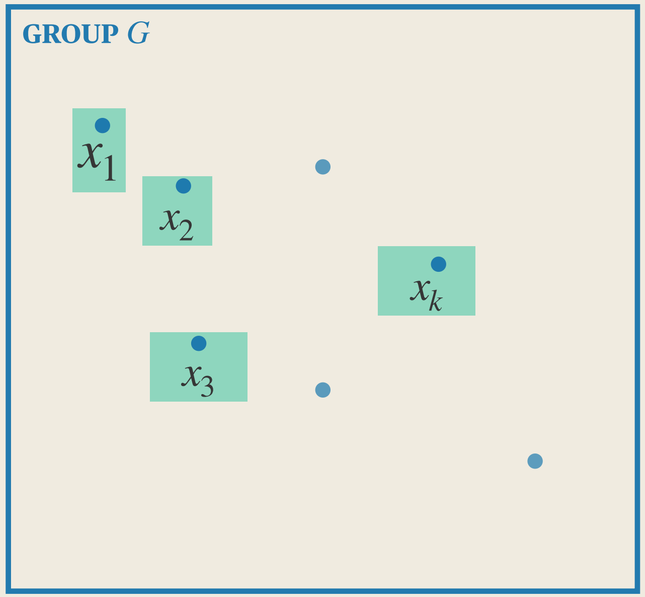


$$f : \mathbb{Z}_8 \to \mathbb{Z}_8$$

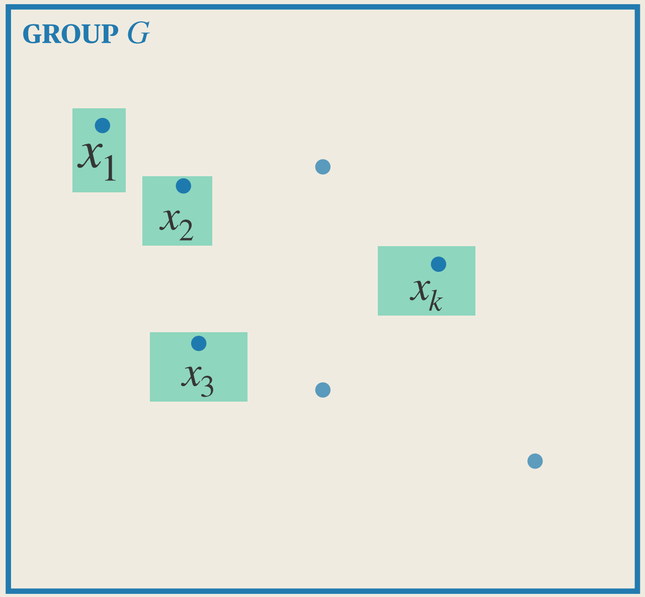
$$f(i) = f(1 + \ldots + 1) = i \cdot f(1)$$



Homomorphism Test

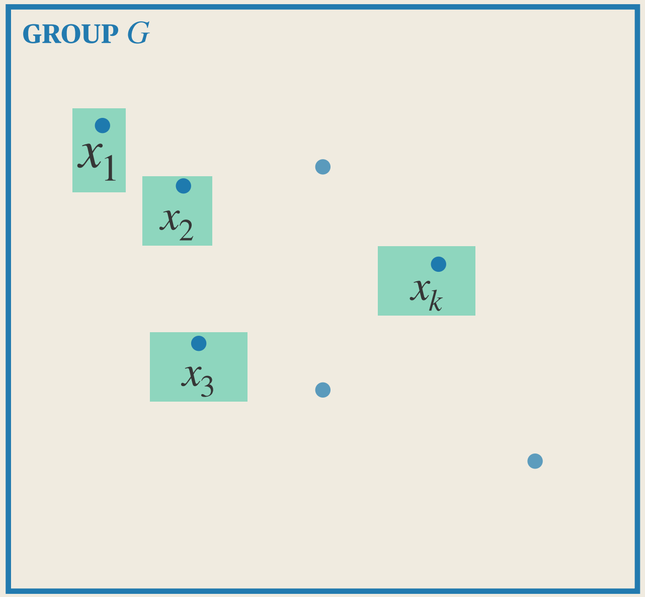


Homomorphism Test

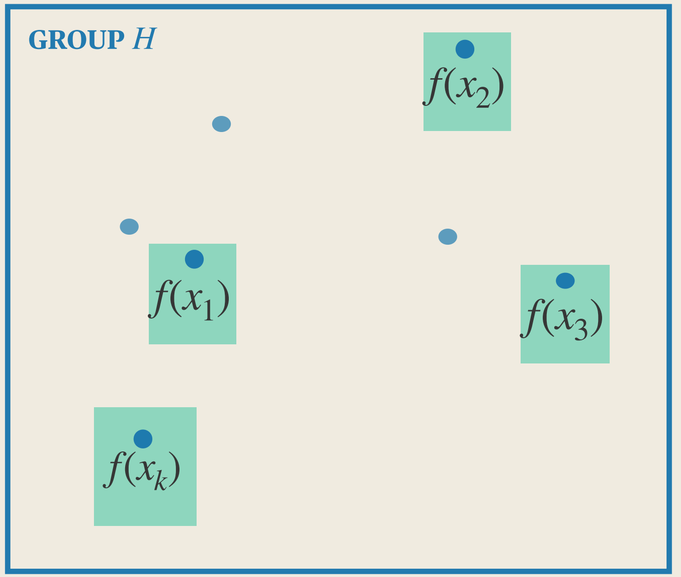


A k-query test queries $f(x_1), \cdots, f(x_k)$

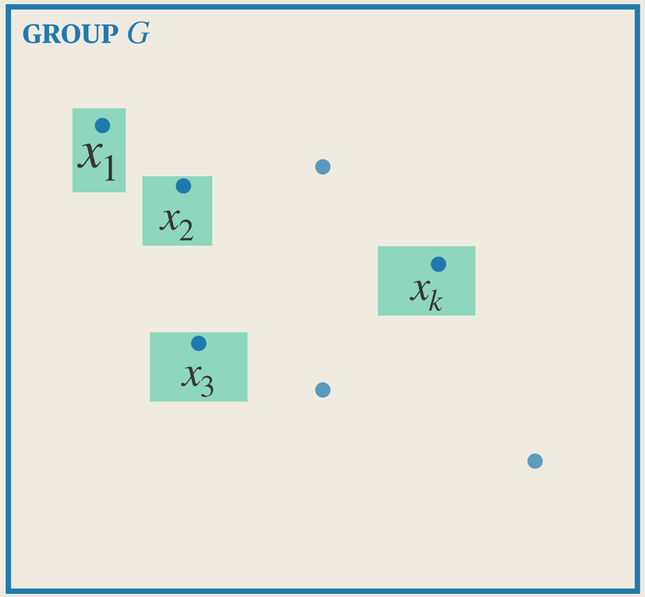
Homomorphism Test



A k-query test queries $f(x_1), \cdots, f(x_k)$

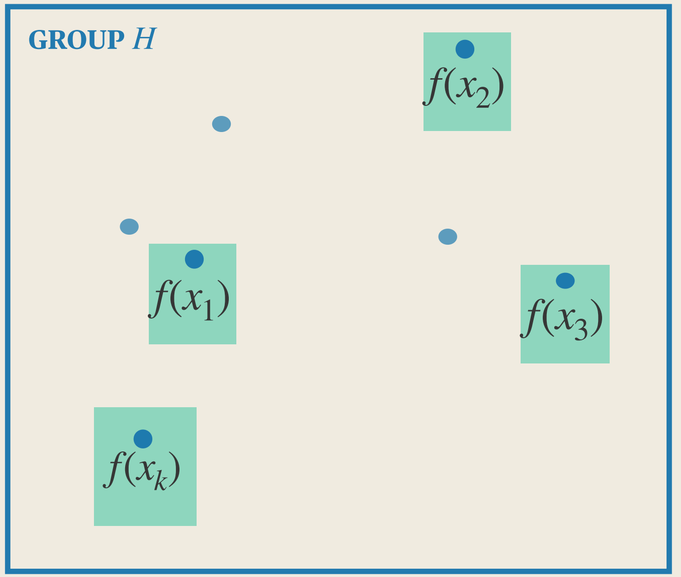


Homomorphism Test



A k-query test queries $f(x_1), \cdots, f(x_k)$

Outputs Yes/No.



Homomorphism Test

A k-query test queries $f(x_1), \cdots, f(x_k)$

Outputs Yes/No.

Homomorphism Test

Completeness

If f is a homomorphism,

outputs YES with probability 1.

A k-query test queries $f(x_1), \cdots, f(x_k)$

Outputs Yes/No.

Homomorphism Test

A k-query test queries $f(x_1), \cdots, f(x_k)$

Outputs Yes/No.

Completeness

If f is a homomorphism,

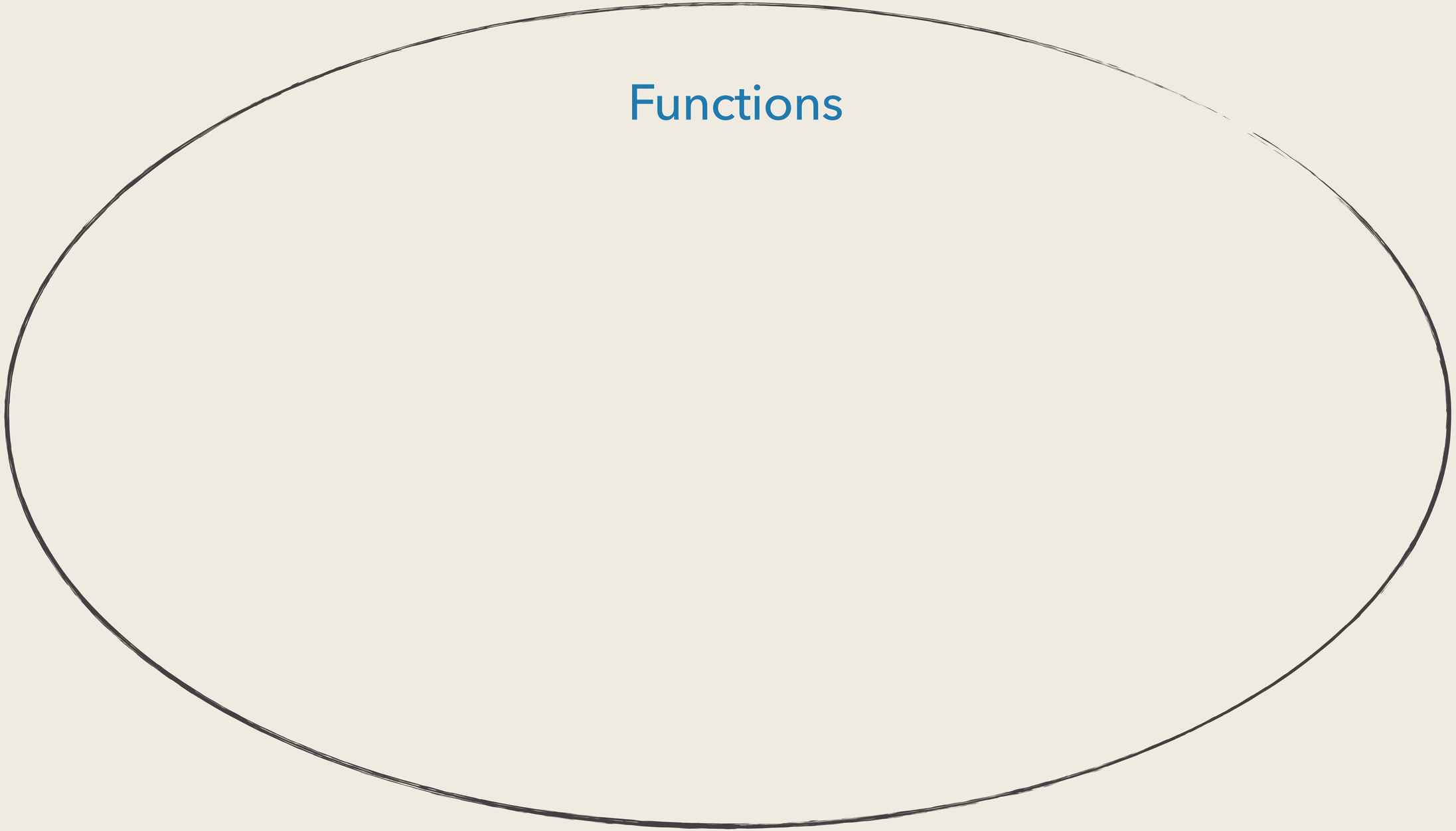
outputs YES with probability 1.

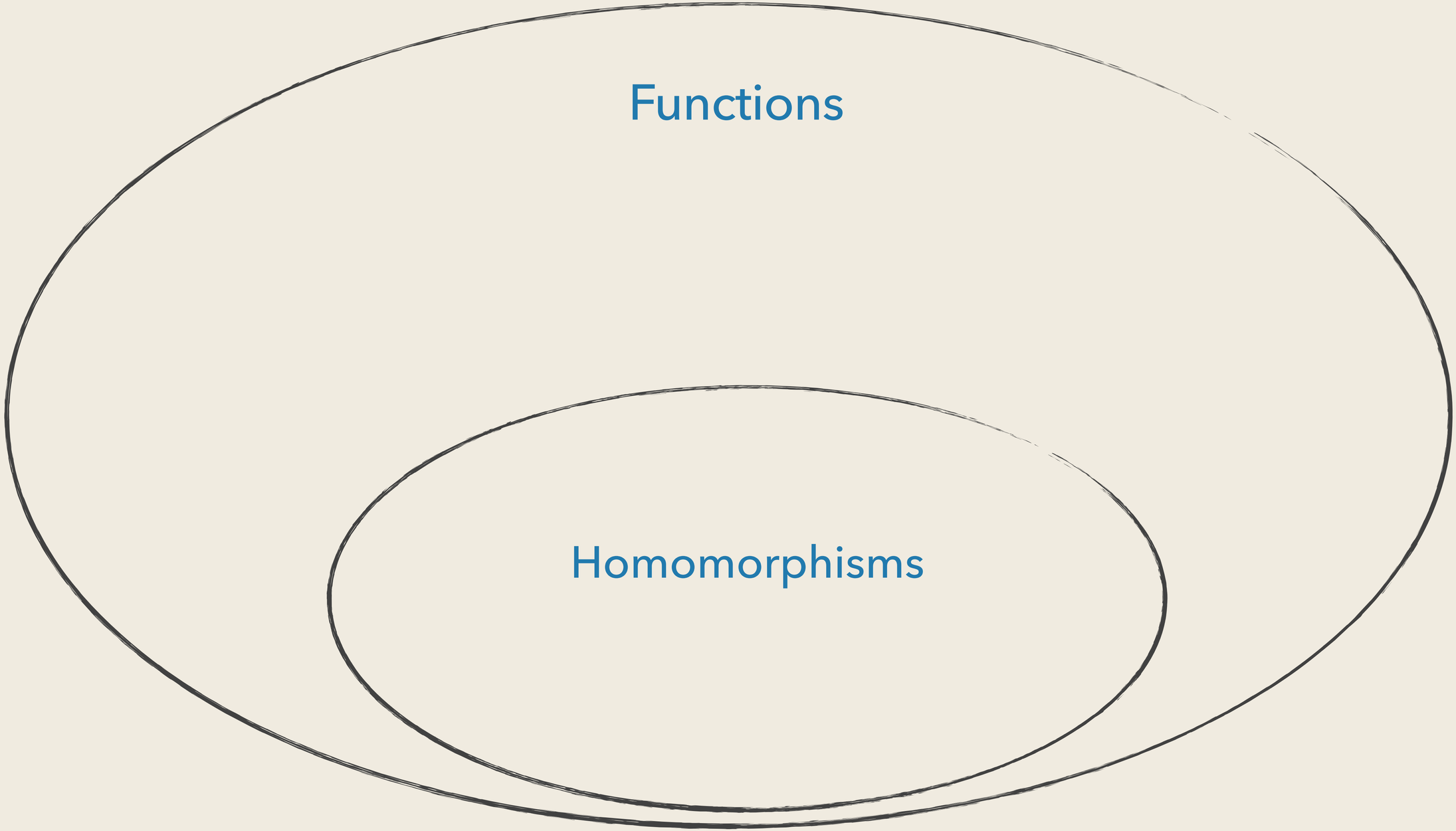
Soundness

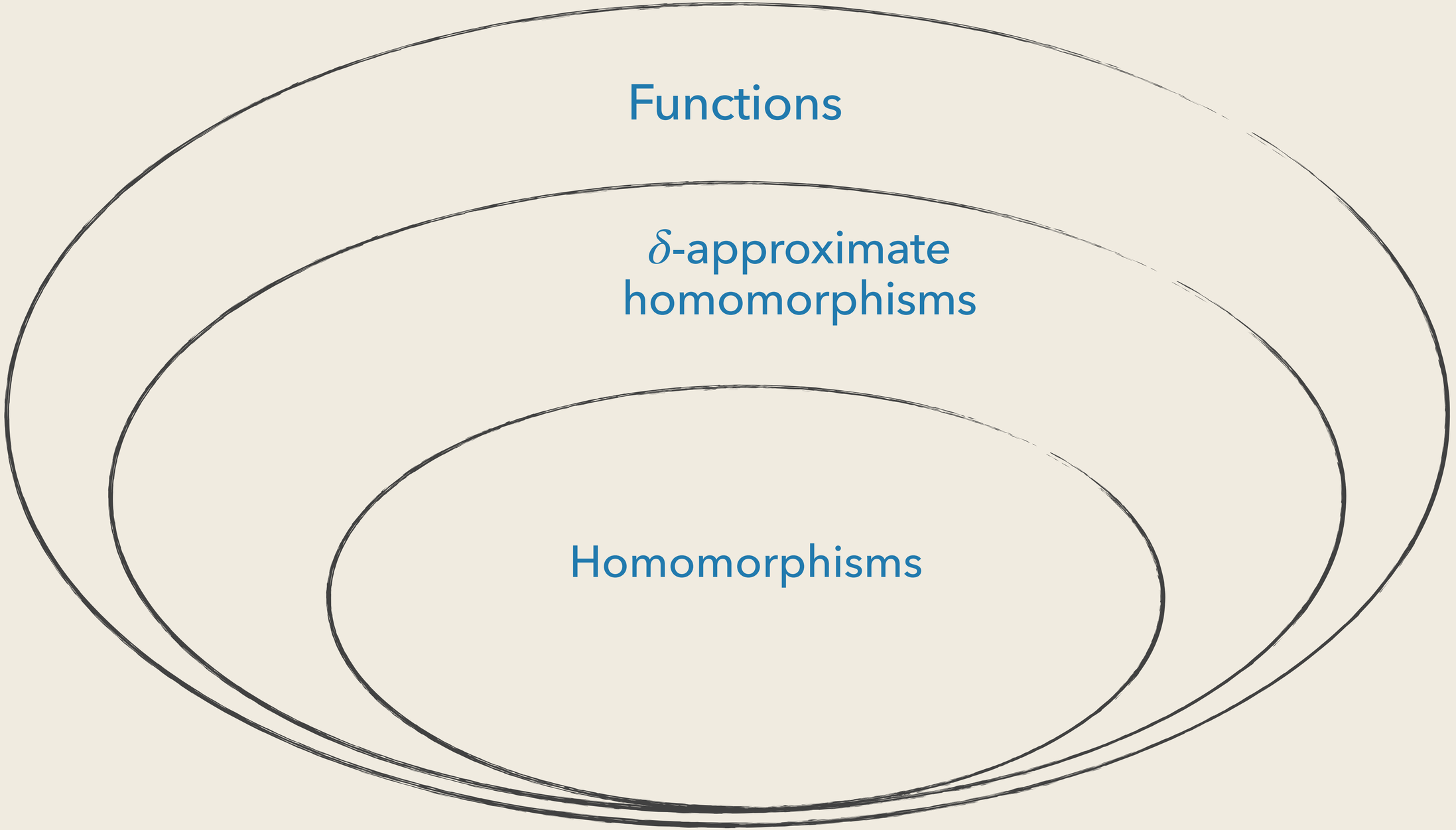
If test outputs YES with probability δ,

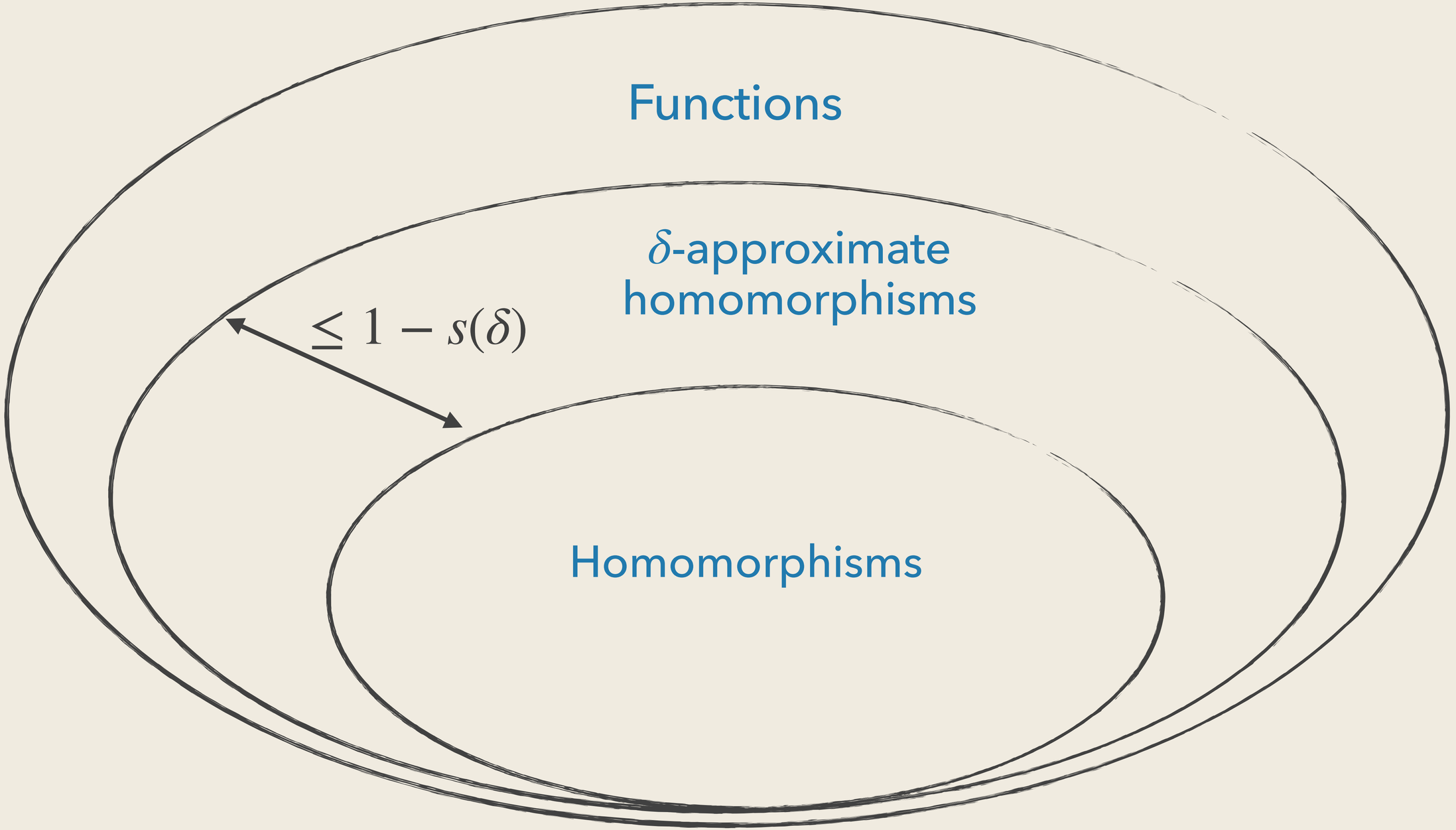
f has $s(\delta)$-agreement with a homomorphism φ

$$\mathrm{agr}(\varphi, f) \;:=\; \Pr_{x \sim G}[f(x) = \varphi(x)] \;\geq\; \mathrm{s}(\delta)\,.$$

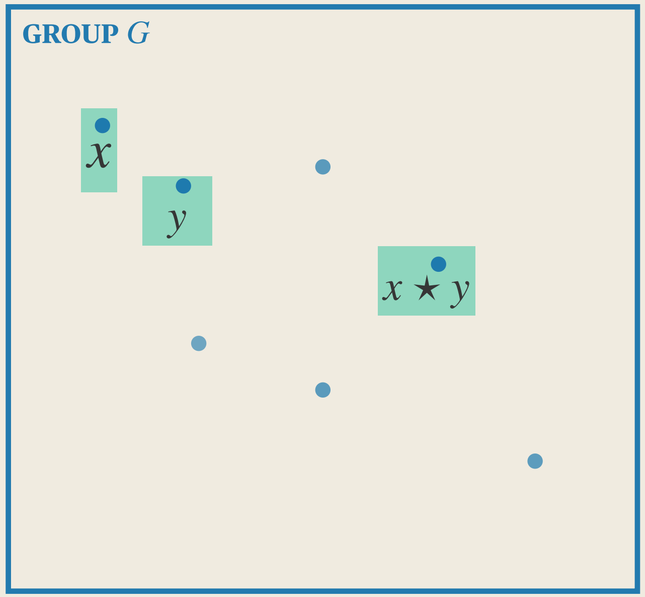

Functions

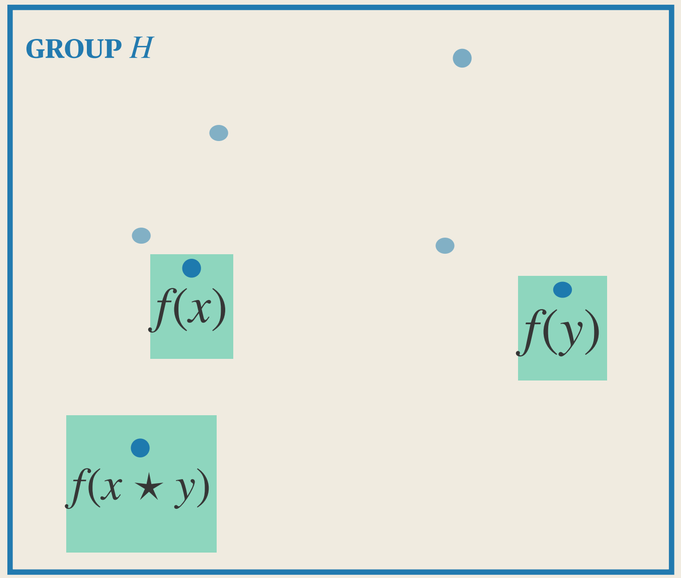

Functions
Homomorphisms


Functions
δ-approximate homomorphisms
Homomorphisms


Functions
δ-approximate homomorphisms
$\leq 1 - s(\delta)$
Homomorphisms

BLR Test: The only known general test

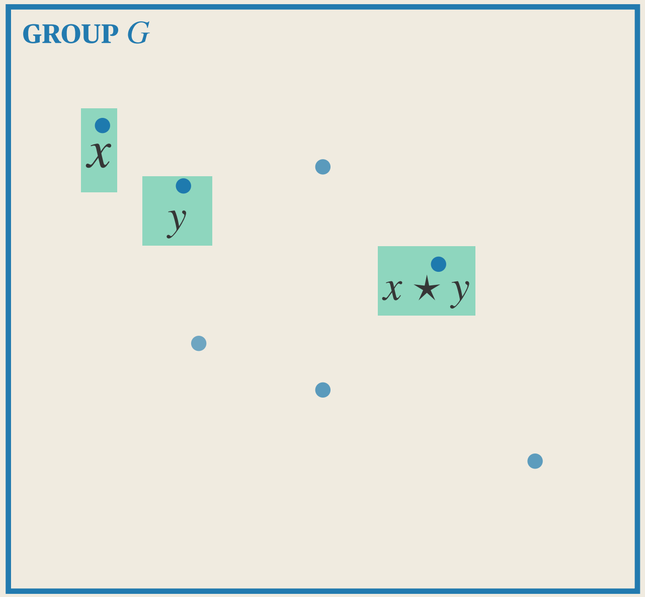


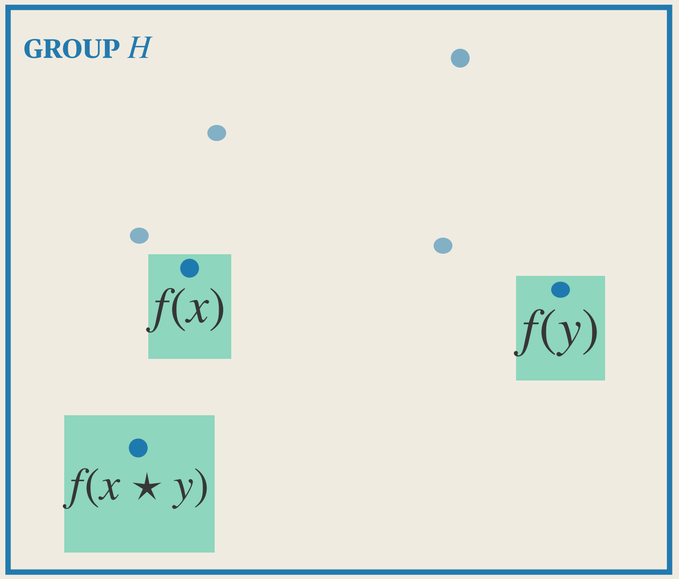


BLR Test: The only known general test

[Blum–Luby–Rubinfeld'90] : One random pair (x, y)

Outputs Yes iff $f(x) \star f(y) \;=\; f(x \star y)$.

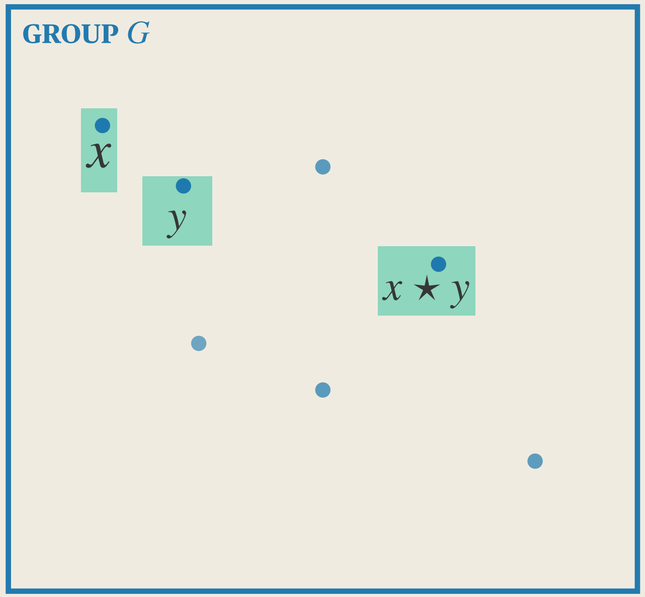




BLR Test: The only known general test

[Blum–Luby–Rubinfeld'90] : One random pair (x, y)

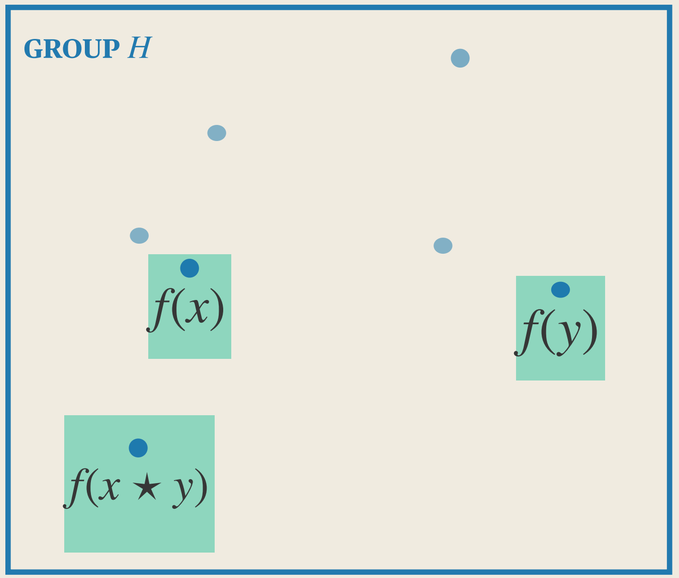
Outputs Yes iff $f(x) \star f(y) = f(x \star y)$.



Completeness

If f is a homomorphism,

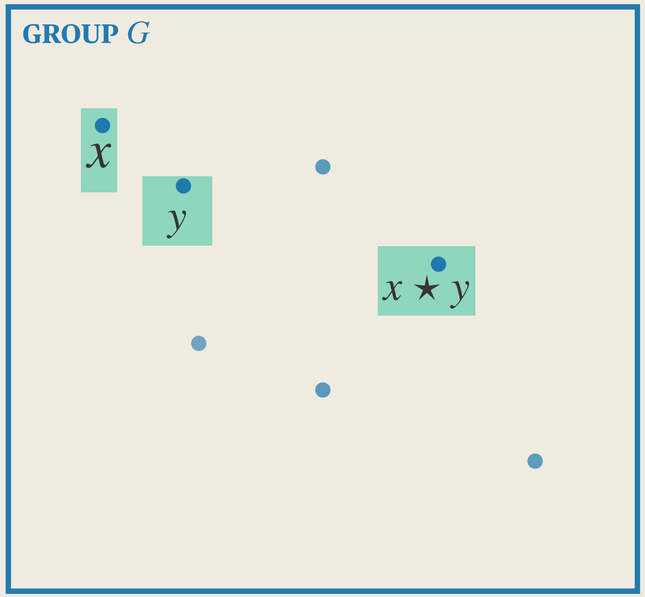
outputs YES with probability 1.



BLR Test: The only known general test

[Blum–Luby–Rubinfeld'90] : One random pair (x, y)

Outputs Yes iff $f(x) \star f(y) \;=\; f(x \star y)$.



Completeness

If f is a homomorphism,

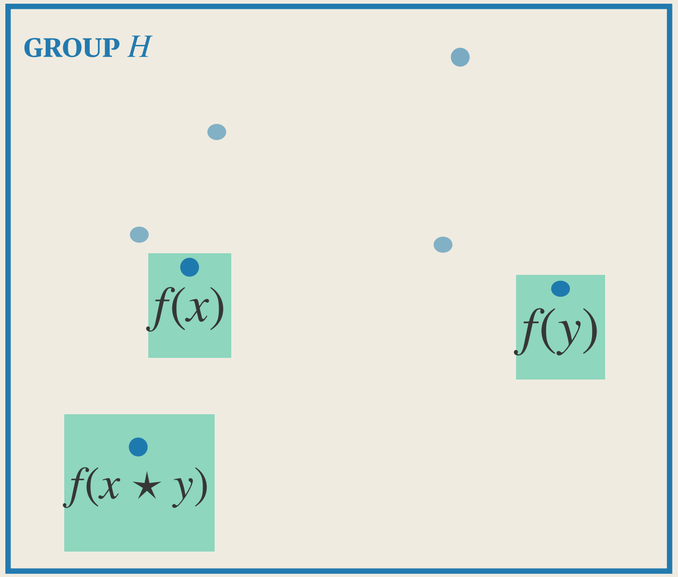
outputs YES with probability 1.

Soundness

If test outputs YES with probability δ,

f has $s(\delta)$-agreement with a homomorphism φ

$$\mathrm{agr}(\varphi, f) \;:=\; \Pr_{x \sim G}[f(x) = \varphi(x)] \;\geq\; \mathrm{s}(\delta)\,.$$



Is the BLR Test sound?

Is the BLR Test sound?

$$\delta_f := \Pr_{x,y\sim G}\left[f(x) \star f(y) \;=\; f(x \star y)\right].$$

Is the BLR Test sound?

$$\delta_f := \Pr_{x,y\sim G}\left[f(x) \star f(y) \;=\; f(x \star y)\right].$$

Does there exists a true homomorphism such that

$$\mathrm{agr}(\varphi, f) \;:=\; \Pr_{x\sim G}[f(x) = \varphi(x)] \;\geq\; \mathrm{s}(\delta_f)\,?$$

BLR poorly understood in the low soundness regime

HIGH SOUNDNESS/UNIQUE DECODING

$\left(\delta > \frac{7}{9}\right)$

LOW SOUNDNESS/LIST DECODING

(ANY NON-TRIVIAL δ)

BLR poorly understood in the low soundness regime

HIGH SOUNDNESS/UNIQUE DECODING

$\left(\delta > \frac{7}{9}\right)$

$f : G \to H$

Any finite G, H

[Blum-Luby-Rubinfeld' 90,
Ben-Or, Coppersmith,Luby, Rubinfeld '04,
Farah'00, Badora-Przebieracz'18]

LOW SOUNDNESS/LIST DECODING

(ANY NON-TRIVIAL δ)

BLR poorly understood in the low soundness regime

HIGH SOUNDNESS/UNIQUE DECODING

$\left(\delta > \frac{7}{9}\right)$

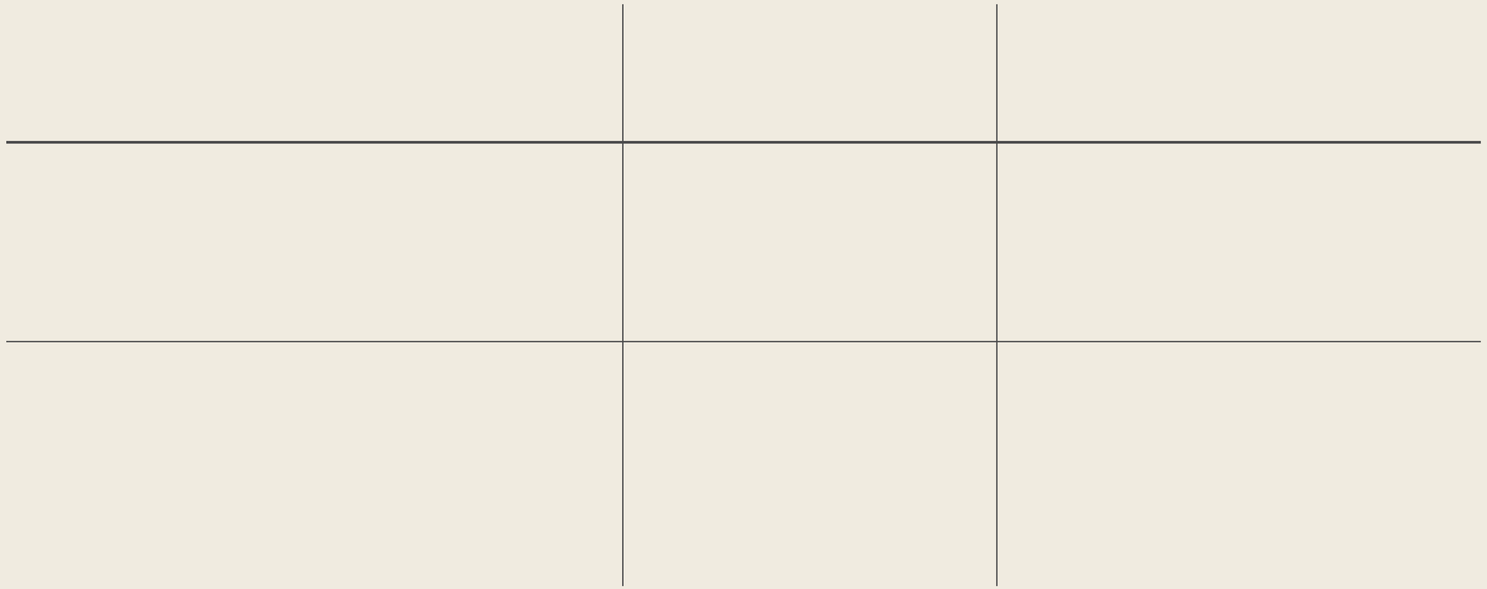
$f : G \to H$

Any finite G, H

[Blum-Luby-Rubinfeld' 90,
Ben-Or, Coppersmith,Luby, Rubinfeld '04,
Farah'00, Badora-Przebieracz'18]

LOW SOUNDNESS/LIST DECODING

(ANY NON-TRIVIAL δ)



BLR poorly understood in the low soundness regime

HIGH SOUNDNESS/UNIQUE DECODING

$\left(\delta > \frac{7}{9}\right)$

$f : G \to H$

Any finite G, H

[Blum-Luby-Rubinfeld' 90,
Ben-Or, Coppersmith,Luby, Rubinfeld '04,
Farah'00, Badora-Przebieracz'18]

LOW SOUNDNESS/LIST DECODING (ANY NON-TRIVIAL δ)

Citation	Setting	Agreement

BLR poorly understood in the low soundness regime

HIGH SOUNDNESS/UNIQUE DECODING

$\left(\delta > \frac{7}{9}\right)$

$f : G \to H$

Any finite G, H

[Blum-Luby-Rubinfeld' 90,
Ben-Or, Coppersmith,Luby, Rubinfeld '04,
Farah'00, Badora-Przebieracz'18]

LOW SOUNDNESS/LIST DECODING (ANY NON-TRIVIAL δ)

Citation	Setting	Agreement
[Bellare-Coppersmith-Hästad-Kiwi Sudan'95, Hästad-Wigderson'02]	$\mathbb{Z}_p^n \to \mathbb{Z}_p$	$\frac{\delta}{p}$

BLR poorly understood in the low soundness regime

HIGH SOUNDNESS/UNIQUE DECODING

$\left(\delta > \frac{7}{9}\right)$

$f : G \to H$

Any finite G, H

[Blum-Luby-Rubinfeld' 90,
Ben-Or, Coppersmith,Luby, Rubinfeld '04,
Farah'00, Badora-Przebieracz'18]

LOW SOUNDNESS/LIST DECODING (ANY NON-TRIVIAL δ)

Citation	Setting	Agreement
[Bellare-Coppersmith-Hästad-Kiwi Sudan'95, Hästad-Wigderson'02]	$\mathbb{Z}_p^n \to \mathbb{Z}_p$	$\frac{\delta}{p}$
[Samrodsnitsky'07, Sanders'12, Gowers-Green-Manners-Tao'23]	$\mathbb{Z}_2^n \to \mathbb{Z}_2^m$	δ^9

BLR poorly understood in the low soundness regime

HIGH SOUNDNESS/UNIQUE DECODING

$\left(\delta > \frac{7}{9}\right)$

$f : G \to H$

Any finite G, H

[Blum-Luby-Rubinfeld' 90,
Ben-Or, Coppersmith,Luby, Rubinfeld '04,
Farah'00, Badora-Przebieracz'18]

LOW SOUNDNESS/LIST DECODING (ANY NON-TRIVIAL δ)

Citation	Setting	Agreement
[Bellare-Coppersmith-Hästad-Kiwi Sudan'95, Hästad-Wigderson'02]	$\mathbb{Z}_p^n \to \mathbb{Z}_p$	$\frac{\delta}{p}$
[Samrodsnitsky'07, Sanders'12, Gowers-Green-Manners-Tao'23]	$\mathbb{Z}_2^n \to \mathbb{Z}_2^m$	δ^9

Same as Freiman-Rusza Problem

BLR generally does not work in low soundness!

BLR generally does not work in low soundness!

[Coppersmith'89]:

BLR fails for cyclic groups in low soundness

For every k, there exists $f : \mathbb{Z}_{3^k} \to \mathbb{Z}_{3^{k-1}}$ with $\delta = \frac{7}{9}$ but

$\mathrm{agr}(f, \varphi) \leq 3^{-(k-1)}$ for every homomorphism φ.

Key Questions

Key Questions

- How do we design tests for a general pair of groups G, H ?

Key Questions

- How do we design tests for a general pair of groups G, H ?
  - Analyzing these tests (beyond Fourier Analysis)?

Key Questions

- How do we design tests for a general pair of groups G, H ?
  - Analyzing these tests (beyond Fourier Analysis)?

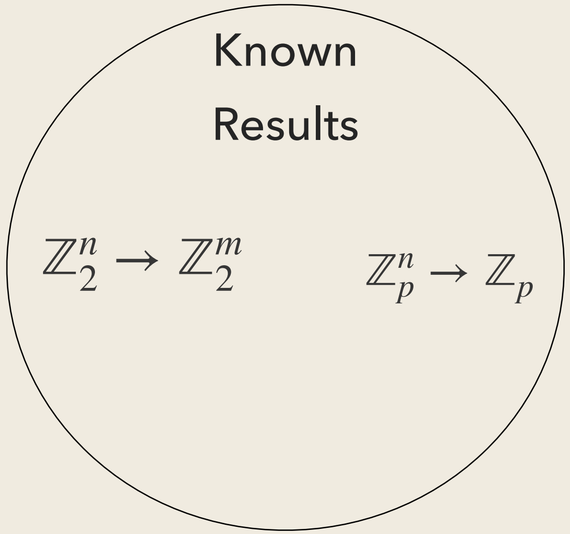
Key Questions

- How do we design tests for a general pair of groups G, H ?
  - Analyzing these tests (beyond Fourier Analysis)?

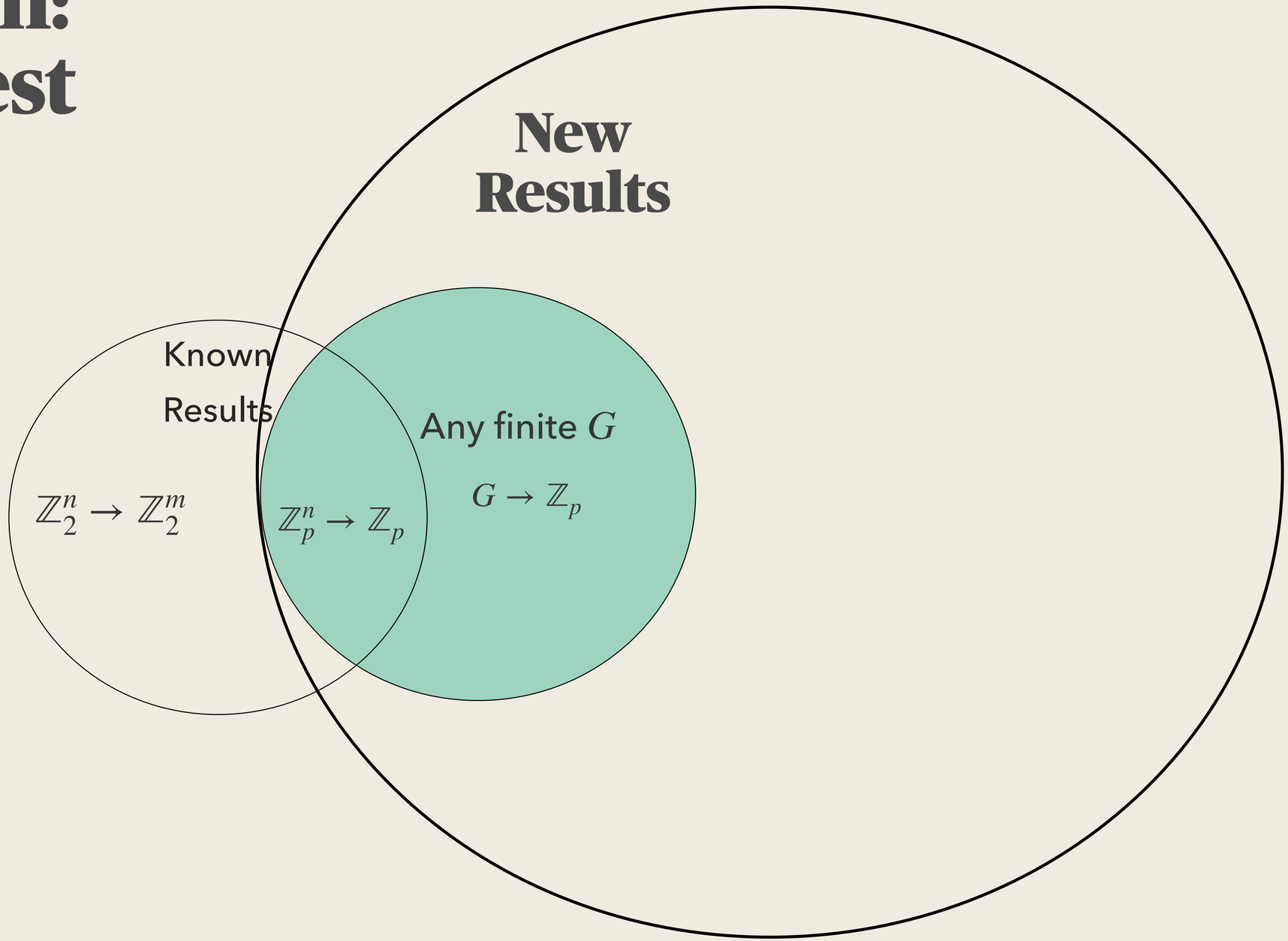
- Trade-off between query complexity and soundness?

Our Contribution:
A new general test

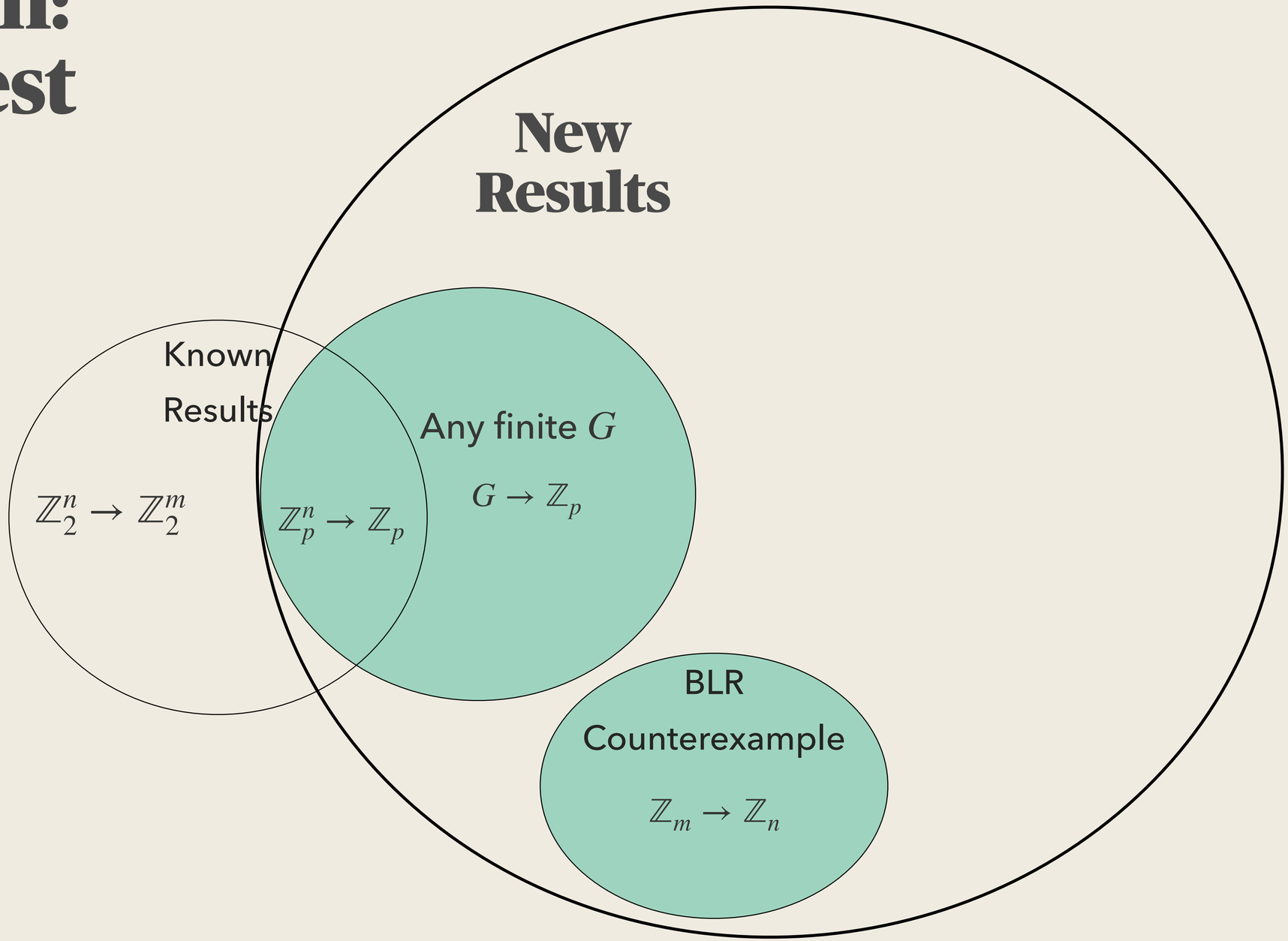
Our Contribution: A new general test



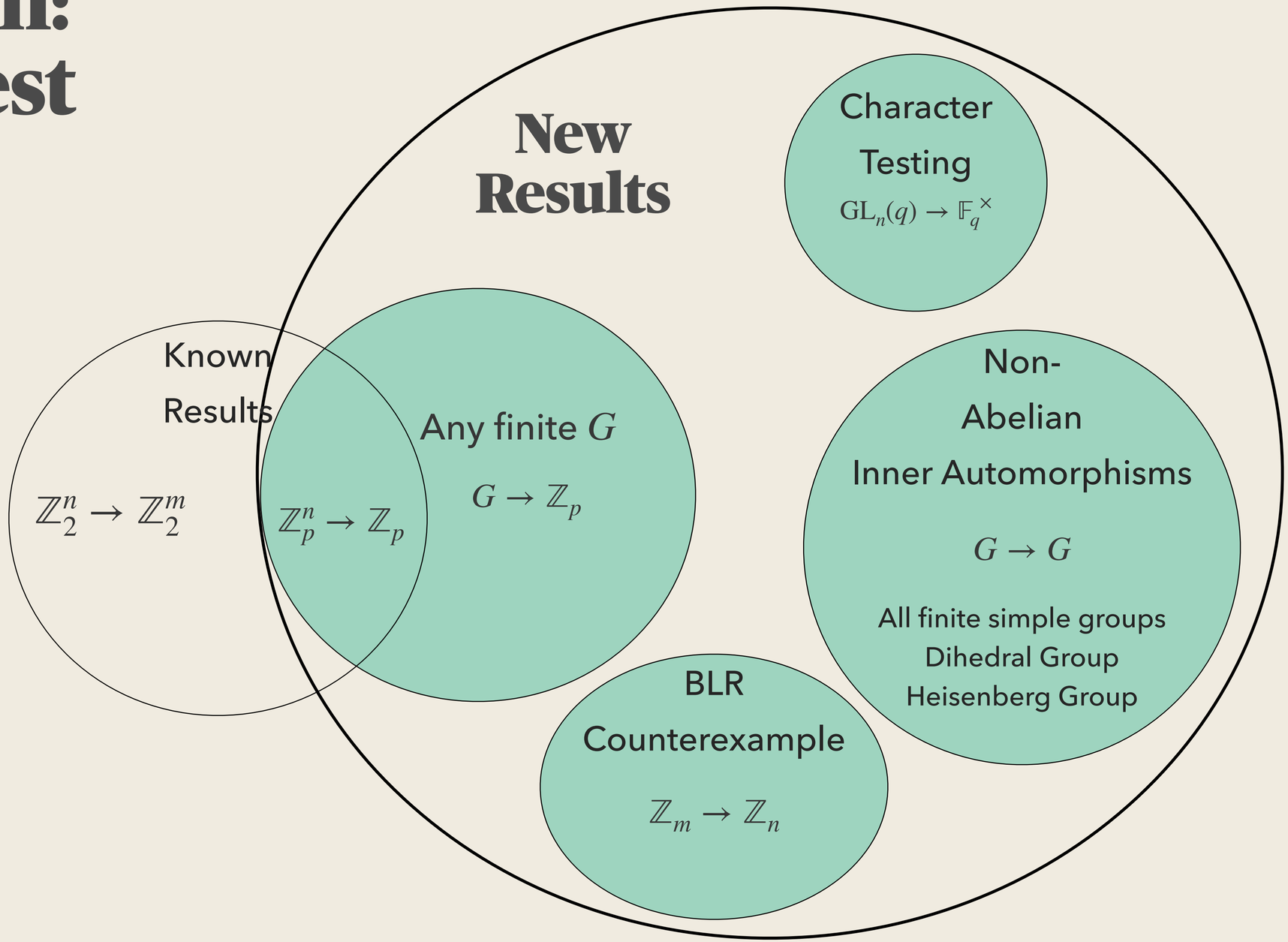
Our Contribution: A new general test



Our Contribution: A new general test

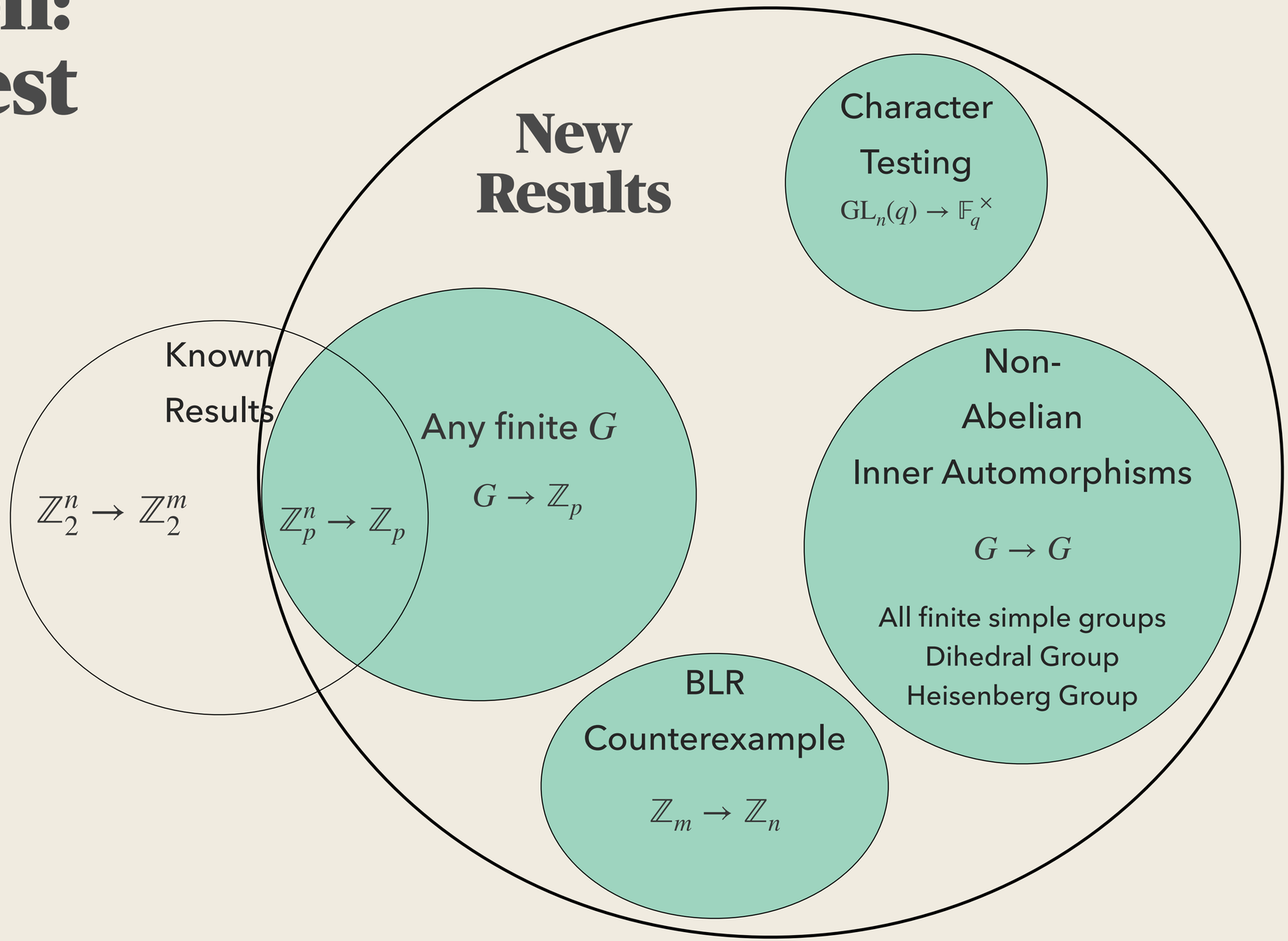


Our Contribution: A new general test



Our Contribution: A new general test

Main New Idea – Queries not uniform (BLR) but a special *group-specific* distribution



General Framework

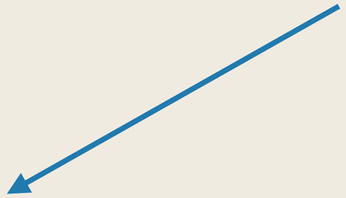
A toy case

A toy case

$$f : \mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$

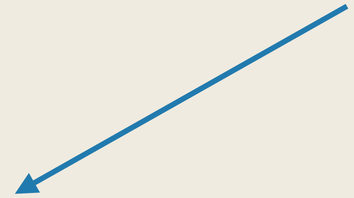
A toy case

$$f : \mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$



A toy case

$$f : \mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$



f is random

$f(0) = 0, \;\; f(x) \sim \{0, \ldots, p^2 - 1\}$

A toy case

$$f : \mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$

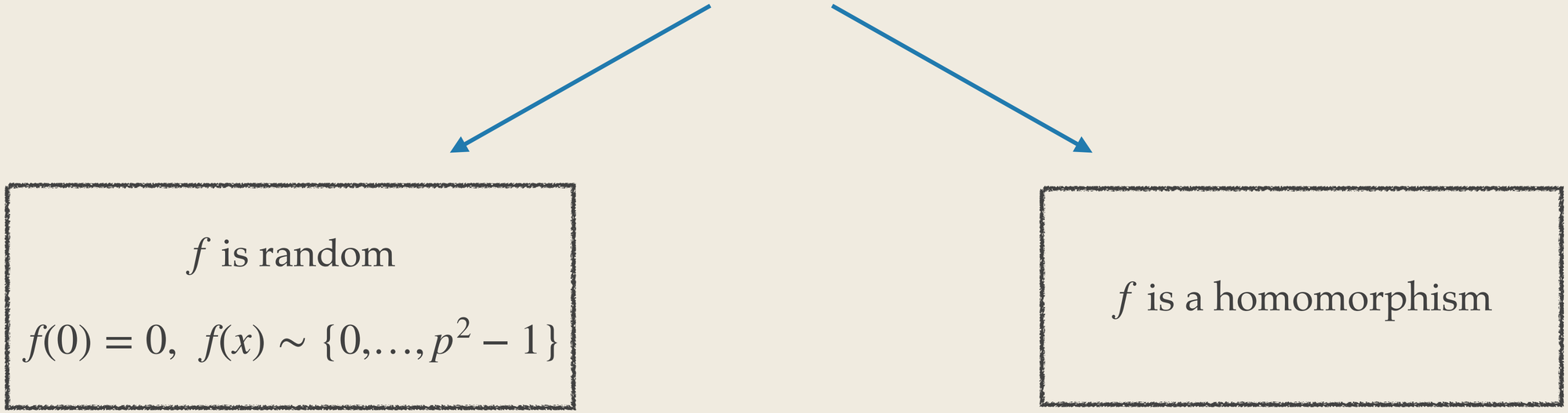


f is random

$f(0) = 0, \;\; f(x) \sim \{0, \ldots, p^2 - 1\}$

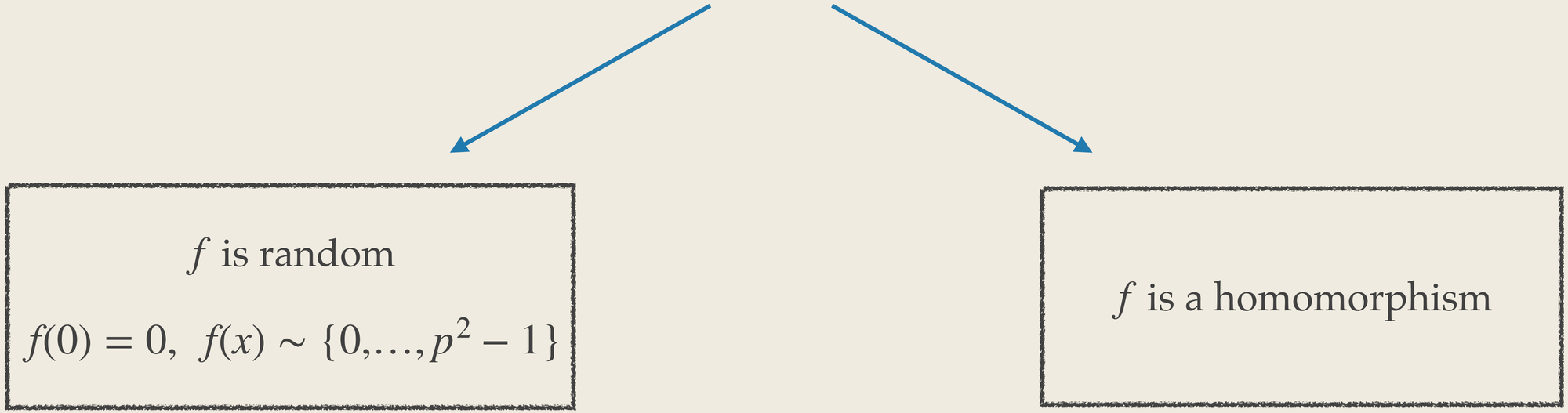
A toy case

$$f : \mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$



A toy case

$$f : \mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$



Decide with 1-query!

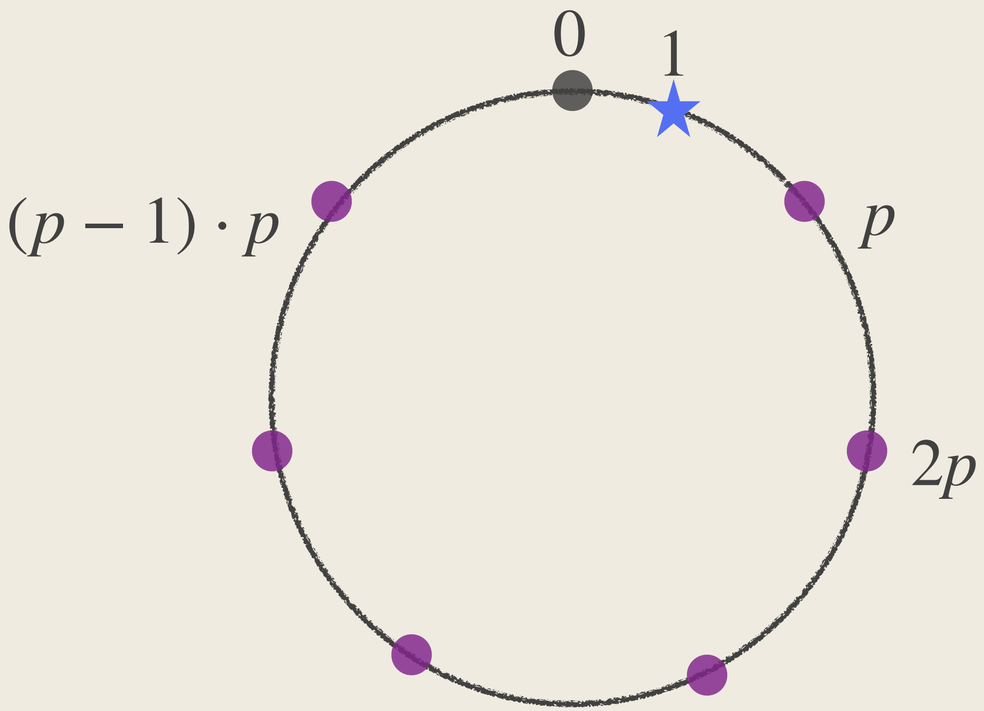
A toy case

A toy case

$$\mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$

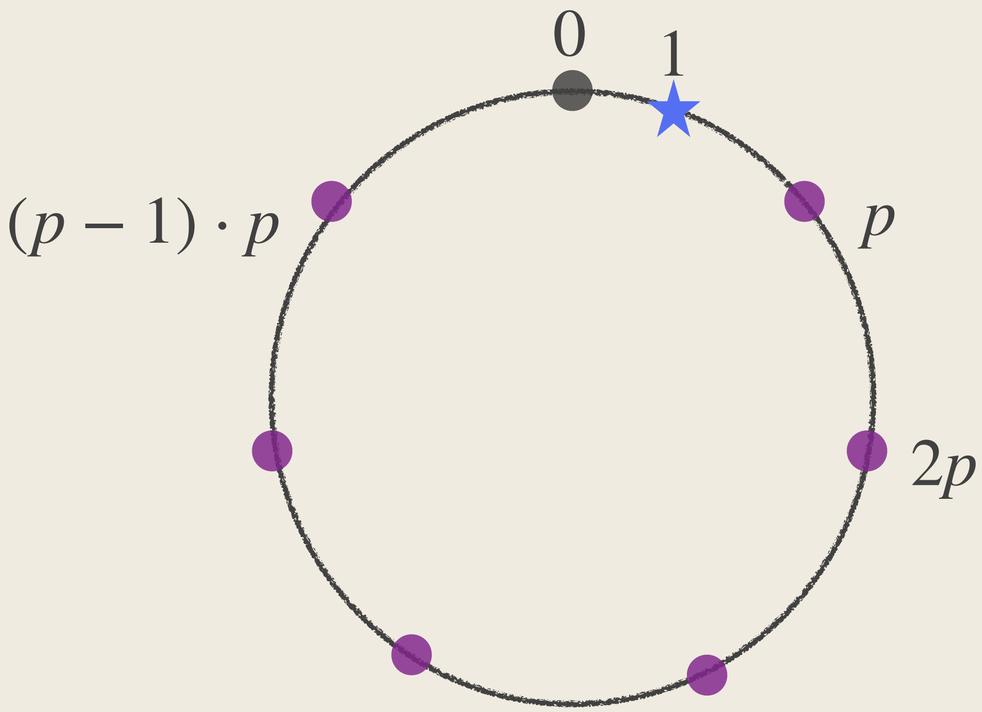
A toy case

$$\mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$



A toy case

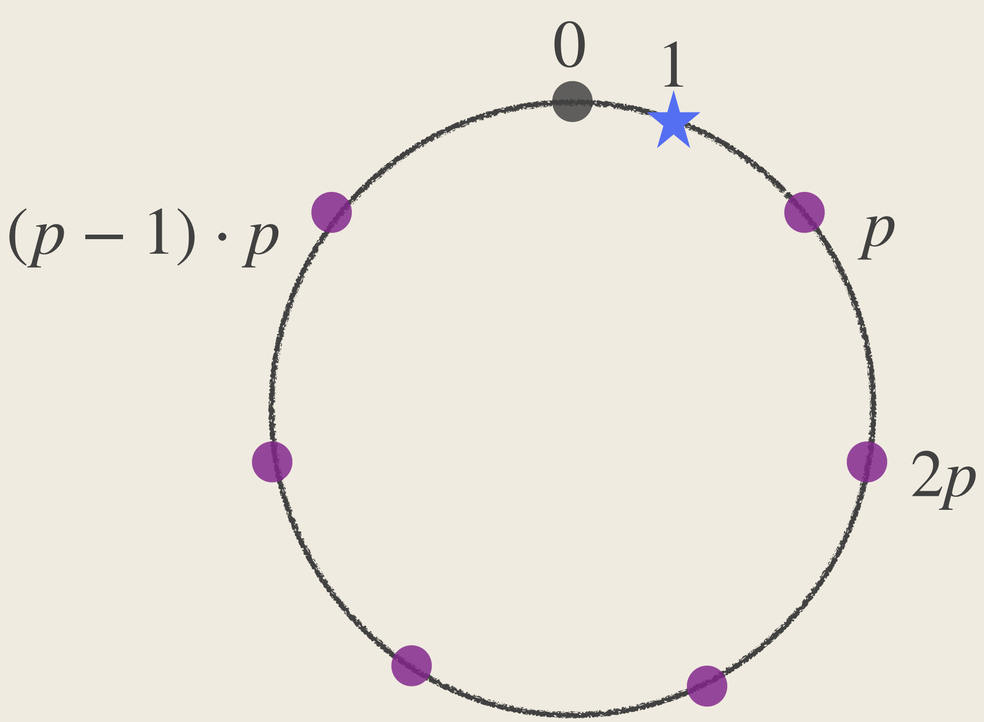
$$\mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$



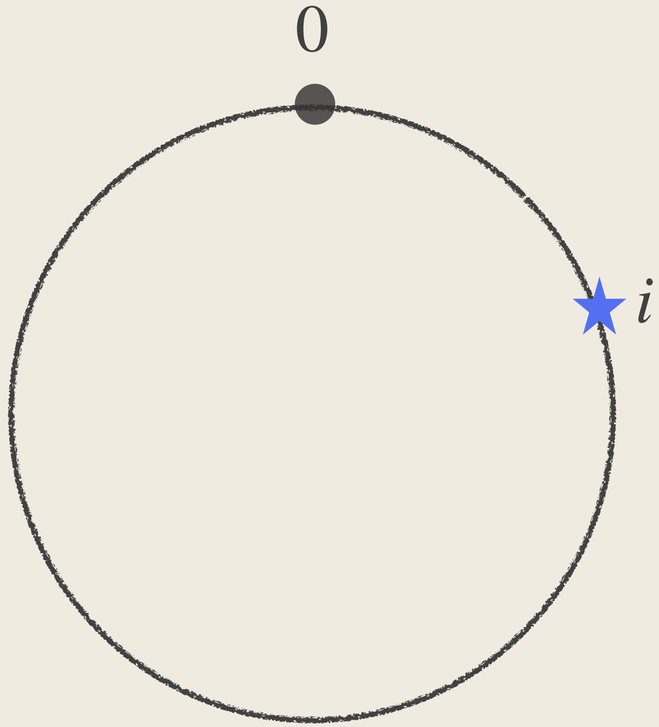
$\phi,\ \phi(1) = i$

A toy case

$$\mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$

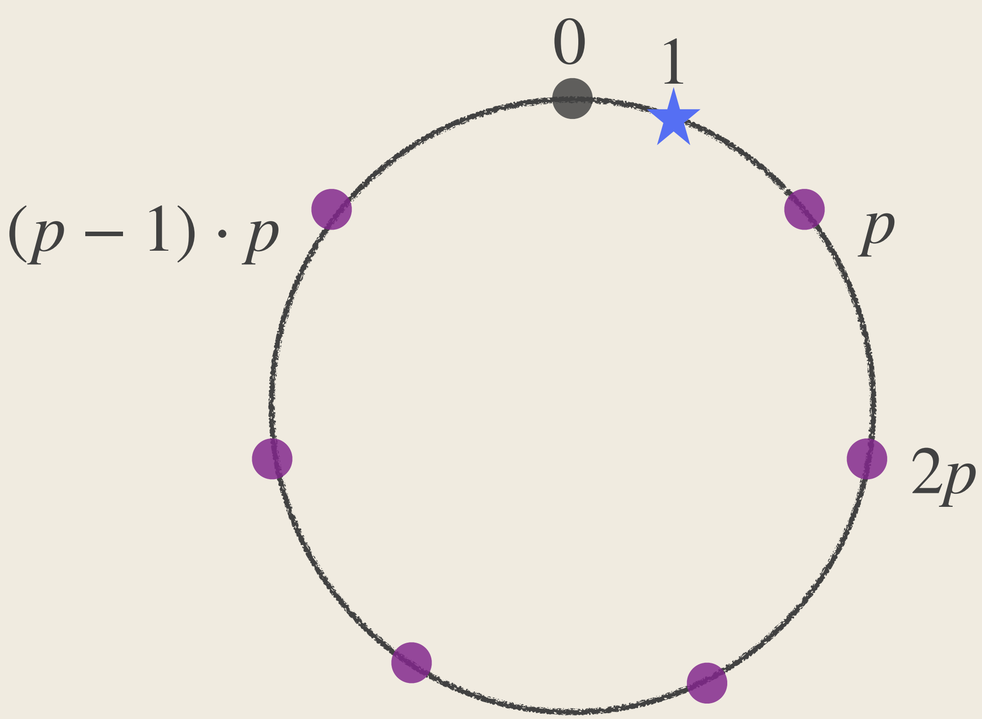


$\phi,\ \phi(1) = i$

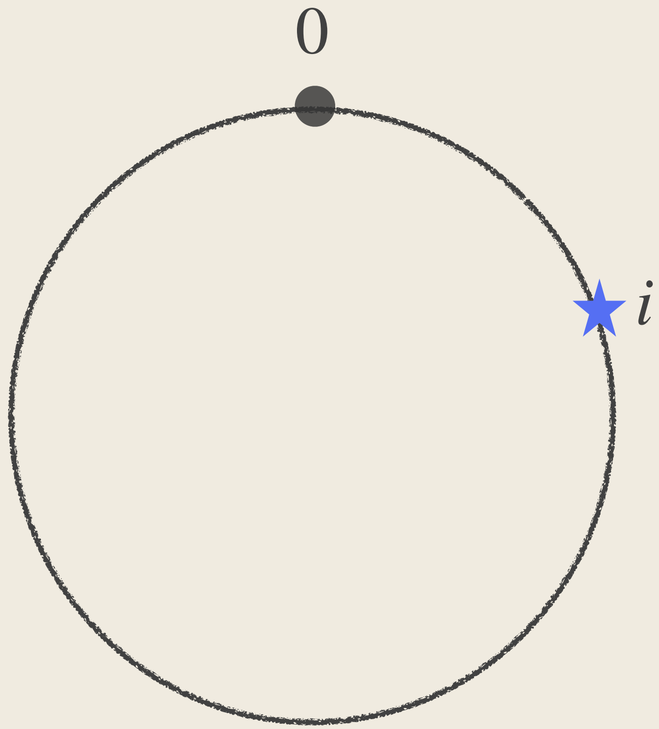


A toy case

$$\mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$

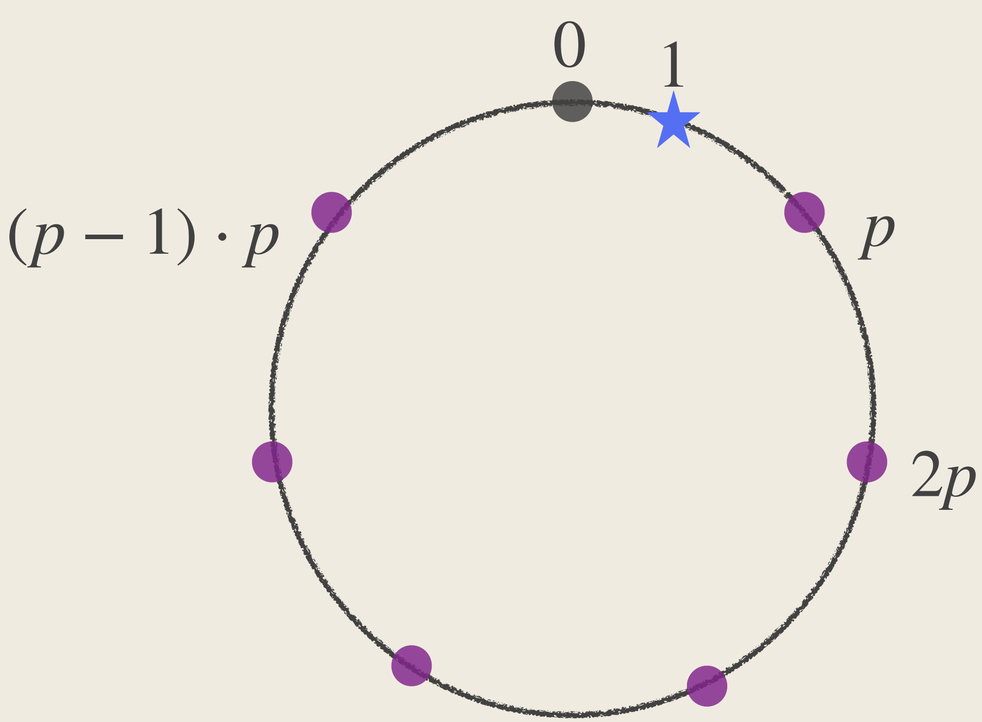


$$\phi,\ \phi(1) = i \xrightarrow[\phi(x) = i \cdot x \mod p^2]{}$$



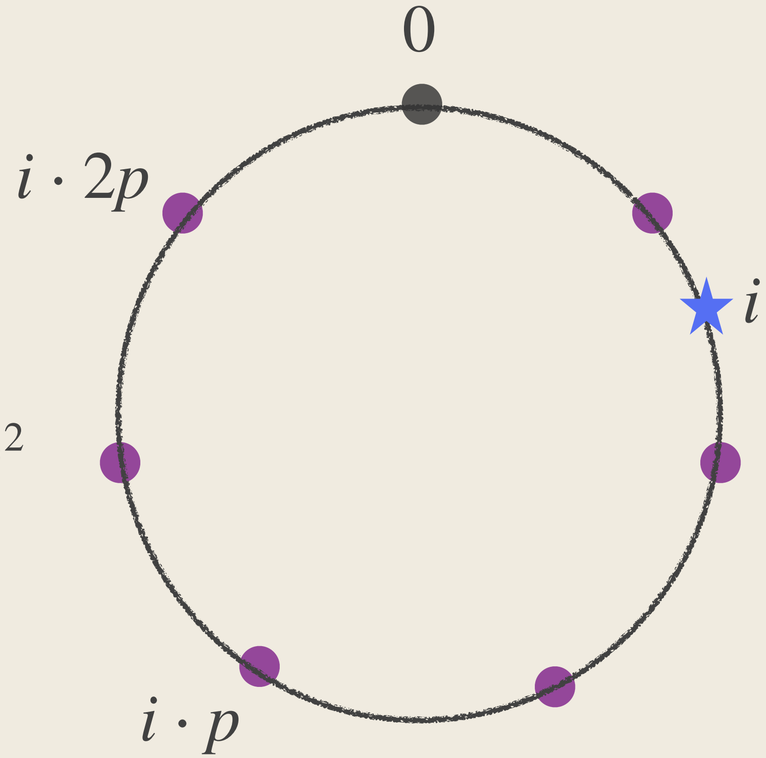
A toy case

$$\mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$



$$\phi,\ \phi(1) = i$$

$$\phi(x) = i \cdot x \mod p^2$$



A toy case

A toy case

- $\phi(1)$ can take all values but $\phi(a \cdot p)$ can only be a multiple of p.

A toy case

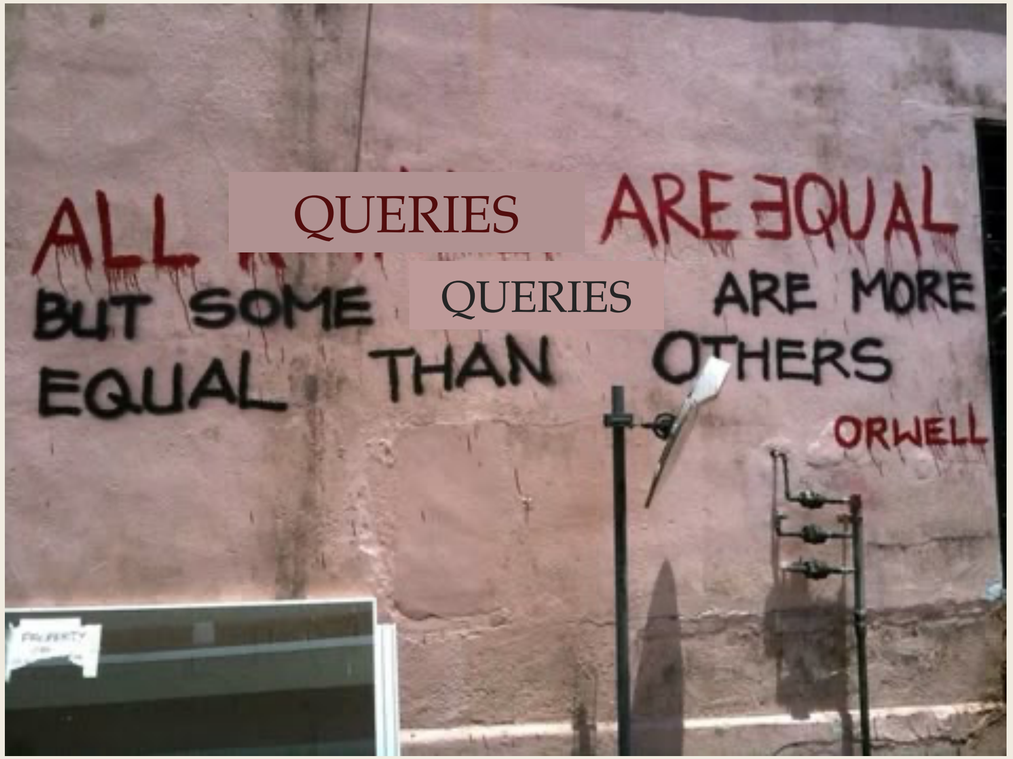
- $\phi(1)$ can take all values but $\phi(a \cdot p)$ can only be a multiple of p.
    - $f(1)$ gives no information, but $f(p)$ does!

A toy case

- $\phi(1)$ can take all values but $\phi(a \cdot p)$ can only be a multiple of p.
    - $f(1)$ gives no information, but $f(p)$ does!

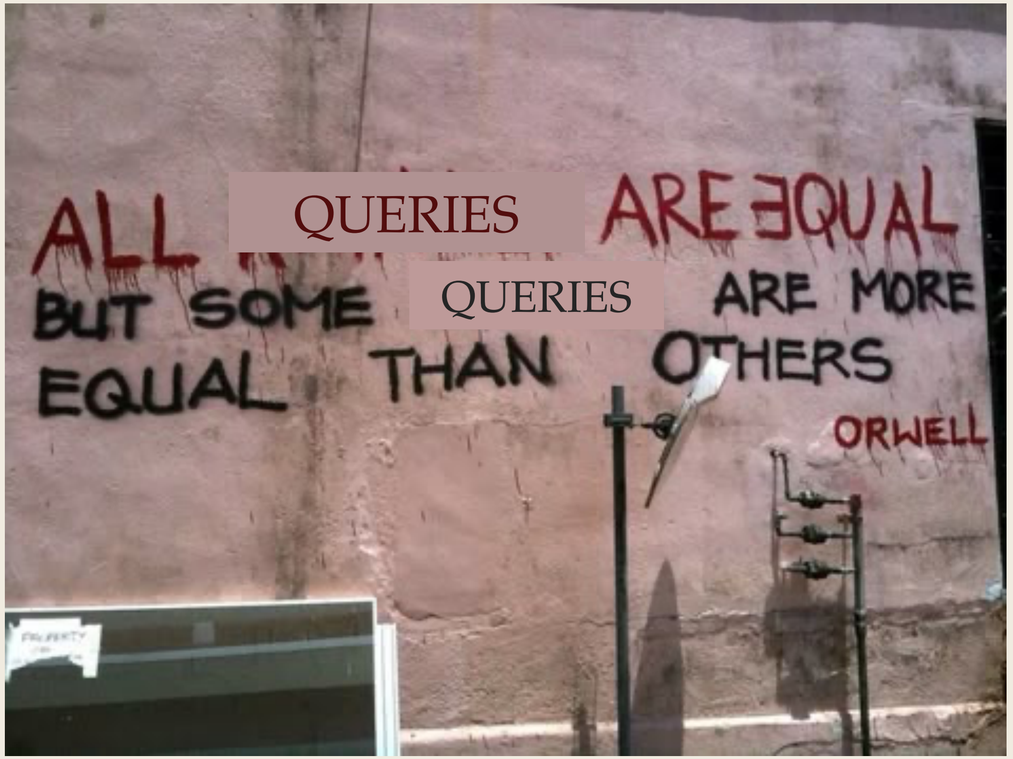
A toy case

- $\phi(1)$ can take all values but $\phi(a \cdot p)$ can only be a multiple of p.
    - $f(1)$ gives no information, but $f(p)$ does!



A toy case

- $\phi(1)$ can take all values but $\phi(a \cdot p)$ can only be a multiple of p.
  - $f(1)$ gives no information, but $f(p)$ does!
  - Takeaway — Need to sample from a "custom" distribution on G^k. How to get one?



Our Approach: Working backwards

Our Approach: Working backwards

- Want to design a test such that for any f :

Our Approach: Working backwards

- Want to design a test such that for any f :

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \geq \mathrm{s}\left(\Pr\left[f \text{ passes Test}\right]\right).$$

Our Approach: Working backwards

- Want to design a test such that for any f :

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \geq \mathrm{s}\left(\Pr\left[f \text{ passes Test}\right]\right).$$

Our Approach: Working backwards

- Want to design a test such that for any f :

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \;\geq\; \mathrm{s}\left(\Pr\left[f \text{ passes Test}\right]\right).$$

- Polynomial Proxy:

Our Approach: Working backwards

- Want to design a test such that for any f :

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f,\varphi) \;\geq\; \mathrm{s}\left(\Pr\left[f \text{ passes Test}\right]\right).$$

- Polynomial Proxy:

$$\sum_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f,\varphi)^k \;\propto\; \Pr\left[f \text{ passes Test}_k\right].$$

The Main Lemma

The Main Lemma

Fix $\vec{x} = (x_1, \ldots, x_k) \in G^k$.

The Main Lemma

Fix $\vec{x} = (x_1, \ldots, x_k) \in G^k$.

$$\Gamma_{\vec{x}} : \ \mathrm{Hom}(G, H) \to H^k$$

The Main Lemma

Fix $\vec{x} = (x_1, \ldots, x_k) \in G^k$.

$$\Gamma_{\vec{x}} : \ \mathrm{Hom}(G, H) \to H^k$$

$$\varphi \mapsto \big(\varphi(x_1), \ldots, \varphi(x_k)\big)$$

The Main Lemma

$$\text{Fix } \vec{x} = (x_1, \ldots, x_k) \in G^k.$$

$$\Gamma_{\vec{x}} : \ \mathrm{Hom}(G, H) \to H^k$$

$$\varphi \mapsto \left(\varphi(x_1), \ldots, \varphi(x_k)\right)$$

$$\sum_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi)^k \ = \ \mathbb{E}_{\vec{x} \in G^k} \left[\mathbb{I}\left[f(\vec{x}) \in \mathrm{Im}(\Gamma_{\vec{x}}) \right] \cdot |\ker(\Gamma_{\vec{x}})| \right].$$

A General Test

Fix $\vec{x} = (x_1, \ldots, x_k) \in G^k$.

$$\Gamma_{\vec{x}} : \ \mathrm{Hom}(G, H) \to H^k$$

$$\varphi \mapsto \big(\varphi(x_1), \ldots, \varphi(x_k)\big)$$

A General Test

Fix $\vec{x} = (x_1, \ldots, x_k) \in G^k$.

$$\Gamma_{\vec{x}} : \ \mathrm{Hom}(G, H) \to H^k$$
$$\varphi \mapsto \big(\varphi(x_1), \ldots, \varphi(x_k)\big)$$

1. Sample $\vec{x} \propto |\ker(\Gamma_{\vec{x}})|$.

A General Test

Fix $\vec{x} = (x_1, \ldots, x_k) \in G^k$.

$$\Gamma_{\vec{x}} : \ \mathrm{Hom}(G, H) \to H^k$$
$$\varphi \mapsto \big(\varphi(x_1), \ldots, \varphi(x_k)\big)$$

1.	Sample $\vec{x} \propto \lvert\ker(\Gamma_{\vec{x}})\rvert$.
2.	Test passes iff $f(\vec{x}) \in \mathrm{Im}(\Gamma_{\vec{x}})$.

A General Test

Fix $\vec{x} = (x_1, \ldots, x_k) \in G^k$.

$$\Gamma_{\vec{x}} : \ \mathrm{Hom}(G, H) \to H^k$$
$$\varphi \mapsto \big(\varphi(x_1), \ldots, \varphi(x_k)\big)$$

1. Sample $\vec{x} \propto |\ker(\Gamma_{\vec{x}})|$.
2. Test passes iff $f(\vec{x}) \in \mathrm{Im}(\Gamma_{\vec{x}})$.

$$|\ker(\Gamma_{\vec{x}})| = \left|\left\{\varphi \mid \varphi(x_1) = \ldots = \varphi(x_k) = 0\right\}\right|$$

Back to our toy example

1. Sample $\vec{x} \propto |\ker(\Gamma_{\vec{x}})|$.

2. Test passes iff $f(\vec{x}) \in \mathrm{Im}(\Gamma_{\vec{x}})$.

Back to our toy example

1. Sample $\vec{x} \propto |\ker(\Gamma_{\vec{x}})|$.
2. Test passes iff $f(\vec{x}) \in \mathrm{Im}(\Gamma_{\vec{x}})$.

$$\mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$

Back to our toy example

1. Sample $\vec{x} \propto |\ker(\Gamma_{\vec{x}})|$.
2. Test passes iff $f(\vec{x}) \in \mathrm{Im}(\Gamma_{\vec{x}})$.

$$\mathbb{Z}_{p^2} \to \mathbb{Z}_{p^2}$$

$$|\ker(\Gamma_x)| = \begin{cases} p & \text{if } x = \alpha \cdot p \\ 1 & \text{otherwise} \end{cases}$$

A General Test

Fix $\vec{x} = (x_1, \ldots, x_k) \in G^k$.

$$\Gamma_{\vec{x}} : \ \mathrm{Hom}(G, H) \to H^k$$
$$\varphi \mapsto \big(\varphi(x_1), \ldots, \varphi(x_k)\big)$$

1.	Sample $\vec{x} \propto \lvert\ker(\Gamma_{\vec{x}})\rvert$.
2.	Test passes iff $f(\vec{x}) \in \mathrm{Im}(\Gamma_{\vec{x}})$.

A General Test

Fix $\vec{x} = (x_1, \ldots, x_k) \in G^k$.

$$\Gamma_{\vec{x}} : \ \mathrm{Hom}(G, H) \to H^k$$
$$\varphi \mapsto \big(\varphi(x_1), \ldots, \varphi(x_k)\big)$$

1. Sample $\vec{x} \propto |\ker(\Gamma_{\vec{x}})|$.
2. Test passes iff $f(\vec{x}) \in \mathrm{Im}(\Gamma_{\vec{x}})$.

$$\sum_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi)^k \ = \ \gamma_k(G, H) \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k\right].$$

Carrying out the analysis

Carrying out the analysis

- Try to bound $\gamma_k(G, H)$ for the given G, H and smallest possible k.

Carrying out the analysis

- Try to bound $\gamma_k(G, H)$ for the given G, H and smallest possible k.

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \geq \frac{\sum_{\varphi} \mathrm{agr}(f, \varphi)^k}{\sum_{\phi} \mathrm{agr}(f, \varphi)^{k-1}}$$

Carrying out the analysis

- Try to bound $\gamma_k(G, H)$ for the given G, H and smallest possible k.

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \geq \frac{\sum_{\varphi} \mathrm{agr}(f, \varphi)^k}{\sum_{\phi} \mathrm{agr}(f, \varphi)^{k-1}}$$

Carrying out the analysis

- Try to bound $\gamma_k(G, H)$ for the given G, H and smallest possible k.

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \geq \frac{\sum_{\varphi} \mathrm{agr}(f, \varphi)^k}{\sum_{\phi} \mathrm{agr}(f, \varphi)^{k-1}}$$

$$\sum_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi)^k = \gamma_k(G, H) \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k\right].$$

$$= \frac{\gamma_k \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k\right]}{\gamma_{k-1} \cdot \Pr\left[f \text{ passes } \mathsf{Test}_{k-1}\right]}$$

Carrying out the analysis

- Try to bound $\gamma_k(G, H)$ for the given G, H and smallest possible k.

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \geq \frac{\sum_{\varphi} \mathrm{agr}(f, \varphi)^k}{\sum_{\phi} \mathrm{agr}(f, \varphi)^{k-1}}$$

$$\sum_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi)^k = \gamma_k(G, H) \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k\right].$$

$$= \frac{\gamma_k \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k\right]}{\gamma_{k-1} \cdot \Pr\left[f \text{ passes } \mathsf{Test}_{k-1}\right]}$$

$$\geq \frac{\gamma_k}{\gamma_{k-1}} \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k\right].$$

Carrying out the analysis

Carrying out the analysis

- Try to bound $\gamma_k(G, H)$ for the given G, H and smallest possible k.

Carrying out the analysis

- Try to bound $\gamma_k(G, H)$ for the given G, H and smallest possible k.

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \geq \frac{\gamma_k}{\gamma_{k-1}} \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k \right].$$

Carrying out the analysis

- Try to bound $\gamma_k(G, H)$ for the given G, H and smallest possible k.

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \geq \frac{\gamma_k}{\gamma_{k-1}} \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k \right].$$

Carrying out the analysis

- Try to bound $\gamma_k(G, H)$ for the given G, H and smallest possible k.

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \geq \frac{\gamma_k}{\gamma_{k-1}} \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k \right].$$



Carrying out the analysis

- Try to bound $\gamma_k(G, H)$ for the given G, H and smallest possible k.

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \geq \frac{\gamma_k}{\gamma_{k-1}} \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k \right].$$

Study a modified term

$$\sum_{\varphi} \left(\mathrm{agr}(f, \varphi) - \tau \right)^k$$

Carrying out the analysis

- Try to bound $\gamma_k(G, H)$ for the given G, H and smallest possible k.

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \geq \frac{\gamma_k}{\gamma_{k-1}} \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k \right].$$

Study a modified term

$$\sum_{\varphi} \left(\mathrm{agr}(f, \varphi) - \tau \right)^k$$

Carrying out the analysis

- Try to bound $\gamma_k(G, H)$ for the given G, H and smallest possible k.

$$\max_{\varphi \in \mathrm{Hom}(G,H)} \mathrm{agr}(f, \varphi) \geq \frac{\gamma_k}{\gamma_{k-1}} \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k \right].$$

Study a modified term

$$\sum_{\varphi} \left(\mathrm{agr}(f, \varphi) - \tau \right)^k$$

Sample $\vec{x} \propto |\ker(\Gamma_{\vec{x}})|$ subject to

$\vec{x} \in \mathscr{D} \subseteq G^k$.

Our Results

Abelian Homomorphism Tests

G	H	Queries	Soundness $[s(\delta)]$

Abelian Homomorphism Tests

G	H	Queries	Soundness $[\mathrm{s}(\delta)]$
$\mathbb{Z}_q^n$	$\mathbb{Z}_q$	3	δ

Abelian Homomorphism Tests

Recovers the known results [BLR, Kiwi'03]

G	H	Queries	Soundness $[\mathrm{s}(\delta)]$
$\mathbb{Z}_q^n$	$\mathbb{Z}_q$	3	δ

Abelian Homomorphism Tests

Recovers the known results [BLR, Kiwi'03]

G	H	Queries	Soundness $[\mathrm{s}(\delta)]$
$\mathbb{Z}_q^n$	$\mathbb{Z}_q$	3	δ
$\mathbb{Z}_{p^r}$	$\mathbb{Z}_{p^s}$	3	$\frac{(p-1)^2}{p^2} \cdot \delta$

Abelian Homomorphism Tests

	G	H	Queries	Soundness $[s(\delta)]$
Recovers the known results [BLR, Kiwi'03]	$\mathbb{Z}_q^n$	$\mathbb{Z}_q$	3	δ
Overcomes Coppersmith's counterexample	$\mathbb{Z}_{p^r}$	$\mathbb{Z}_{p^s}$	3	$\frac{(p-1)^2}{p^2} \cdot \delta$

Abelian Homomorphism Tests

	G	H	Queries	Soundness $[s(\delta)]$
Recovers the known results [BLR, Kiwi'03]	$\mathbb{Z}_q^n$	$\mathbb{Z}_q$	3	δ
Overcomes Coppersmith's counterexample	$\mathbb{Z}_{p^r}$	$\mathbb{Z}_{p^s}$	3	$\frac{(p-1)^2}{p^2} \cdot \delta$
	$\mathbb{Z}_{p^r}$	Abelian group of p-rank $\leq t$	$t+2$	$\frac{(p-1)^2}{p^2} \cdot \delta$

Abelian Homomorphism Tests

Recovers the known results [BLR, Kiwi'03]

Overcomes Coppersmith's counterexample

G	H	Queries	Soundness $[s(\delta)]$
$\mathbb{Z}_q^n$	$\mathbb{Z}_q$	3	δ
$\mathbb{Z}_{p^r}$	$\mathbb{Z}_{p^s}$	3	$\frac{(p-1)^2}{p^2} \cdot \delta$
$\mathbb{Z}_{p^r}$	Abelian group of p-rank $\leq t$	$t+2$	$\frac{(p-1)^2}{p^2} \cdot \delta$
$\mathbb{Z}_n$	$\mathbb{Z}_m$	4	$\zeta(2)^{-2} \cdot \delta$

Abelian Homomorphism Tests

Recovers the known results [BLR, Kiwi'03]

Overcomes Coppersmith's counterexample

G	H	Queries	Soundness $[s(\delta)]$
$\mathbb{Z}_q^n$	$\mathbb{Z}_q$	3	δ
$\mathbb{Z}_{p^r}$	$\mathbb{Z}_{p^s}$	3	$\frac{(p-1)^2}{p^2} \cdot \delta$
$\mathbb{Z}_{p^r}$	Abelian group of p-rank $\leq t$	$t+2$	$\frac{(p-1)^2}{p^2} \cdot \delta$
$\mathbb{Z}_n$	$\mathbb{Z}_m$	4	$\zeta(2)^{-2} \cdot \delta$
$\mathbb{Z}_q$	$\mathbb{Z}_q^n$	2	$\frac{q-1}{q} \cdot \delta$

Non-Abelian Automorphism Tests

Non-Abelian Automorphism Tests

- Automorphism $\phi : G \to G$, bijection.

Non-Abelian Automorphism Tests

- Automorphism $\phi : G \to G$, bijection.
  - Inner Automorphisms – $\phi_g(x) = gxg^{-1}$ ("Conjugation Equivalence")

Non-Abelian Automorphism Tests

- Automorphism $\phi : G \to G$, bijection.
  - Inner Automorphisms – $\phi_g(x) = gxg^{-1}$ (“Conjugation Equivalence”)

Theorem

Non-Abelian Automorphism Tests

- Automorphism $\phi : G \to G$, bijection.
  - Inner Automorphisms – $\phi_g(x) = gxg^{-1}$ ("Conjugation Equivalence")

Theorem

- Automorphisms: 3-query tests for the dihedral group and the symmetric group.

Non-Abelian Automorphism Tests

- Automorphism $\phi : G \to G$, bijection.
  - Inner Automorphisms – $\phi_g(x) = gxg^{-1}$ ("Conjugation Equivalence")

Theorem

- Automorphisms: 3-query tests for the dihedral group and the symmetric group.
- Inner Automorphisms: 4-query tests for every *non-abelian finite simple group*, *quasirandom groups*, and '*extraspecial groups*'.

Non-Abelian Automorphism Tests

- Automorphism $\phi : G \to G$, bijection.
  - Inner Automorphisms – $\phi_g(x) = gxg^{-1}$ ("Conjugation Equivalence")

Theorem

- Automorphisms: 3-query tests for the dihedral group and the symmetric group.
- Inner Automorphisms: 4-query tests for every *non-abelian finite simple group*, *quasirandom groups*, and '*extraspecial groups*'.
  - *Extraspecial groups* – generalize *Heisenberg groups,* often pop up in quantum.

Key Technical Work: Bounding $\gamma_k(G, H)$

Key Technical Work: Bounding $\gamma_k(G, H)$

$$\gamma_k(G,H) = \sum_{\phi \in \mathrm{Hom(G,H)}} |\ker(\phi)|^k$$

Key Technical Work: Bounding $\gamma_k(G, H)$

$$\gamma_k(G,H) = \sum_{\phi \in \mathrm{Hom(G,H)}} |\ker(\phi)|^k$$

Key Technical Work: Bounding $\gamma_k(G, H)$

$$\gamma_k(G,H) = \sum_{\phi \in \mathrm{Hom(G,H)}} |\ker(\phi)|^k$$

- Inner automorphism testing – connections to “analogs of Witten-Zeta function”

Key Technical Work: Bounding $\gamma_k(G, H)$

$$\gamma_k(G,H) \;=\; \sum_{\phi \in \mathrm{Hom(G,H)}} |\ker(\phi)|^k$$

- Inner automorphism testing – connections to “analogs of Witten-Zeta function”
  - Non-abelian analog of $\gamma_k(G, G) \;=\; \sum_C |C|^{1-k}$

Key Technical Work: Bounding $\gamma_k(G, H)$

$$\gamma_k(G,H) \;=\; \sum_{\phi \in \mathrm{Hom(G,H)}} |\ker(\phi)|^k$$

- Inner automorphism testing – connections to “analogs of Witten-Zeta function”
  - Non-abelian analog of $\gamma_k(G, G) \;=\; \sum_C |C|^{1-k}$
- Use known deep results on character bounds, sizes of conjugacy classes.

Key Technical Work: Bounding $\gamma_k(G, H)$

$$\gamma_k(G,H) = \sum_{\phi \in \mathrm{Hom}(G,H)} |\ker(\phi)|^k$$

- Inner automorphism testing – connections to "analogs of Witten-Zeta function"
  - Non-abelian analog of $\gamma_k(G,G) = \sum_C |C|^{1-k}$
- Use known deep results on character bounds, sizes of conjugacy classes.
  - Give a "direct proof" using only *quasirandomness*.

Feature 1: Query vs Soundness

G	H	Queries (any $k \geq 1$)	Soundness $[\mathrm{s}(\delta)]$
$\mathbb{Z}_q^n$	$\mathbb{Z}_q$	Odd, $2+k$	$O(\delta)^{\frac{1}{k}}$
$\mathbb{Z}_{p^r}$	$\mathbb{Z}_{p^s}$	$2+k$	
$\mathbb{Z}_{p^r}$	Abelian group of p-rank $\leq t$	$t+1+k$	
$\mathbb{Z}_n$	$\mathbb{Z}_m$	$3+k$	
$\mathbb{Z}_q$	$\mathbb{Z}_q^n$	$1+k$	

Feature 2: List Decoding Bounds

- Want a bound on $L_\varepsilon = \{\varphi \mid \mathrm{agr}(f, \phi) \geq \varepsilon\}$ for any f.

Feature 2: List Decoding Bounds

- Want a bound on $L_\varepsilon = \{\varphi \mid \mathrm{agr}(f, \phi) \geq \varepsilon\}$ for any f.

$$\sum_{\varphi} \mathrm{agr}(f, \varphi)^k \;=\; \gamma_k \cdot \Pr\left[f \text{ passes } \mathsf{Test}_k\right] \;\leq\; \gamma_k \,.$$

Feature 2: List Decoding Bounds

Feature 2: List Decoding Bounds

Our Framework

k query test $\implies$

Number of homomorphisms with ϵ-agreement $\leq$ $O(\varepsilon^{1-k})$.

Feature 2: List Decoding Bounds

Our Framework

k query test $\implies$

Number of homomorphisms with ϵ-agreement $\leq O(\varepsilon^{1-k})$.

$|L_\varepsilon| \leq O(\varepsilon^{-105})$

[Dinur, Grigorescu, Kopparty, Sudan'08]

G, H cyclic groups

(prime power order)

$|L_\varepsilon| \leq 2\varepsilon^{-2}$

[Our Framework]

Feature 3: Lifting

Feature 3: Lifting

$$\mathrm{Hom}(G, H) \cong \mathrm{Hom}(A, B).$$

The framework can use $\gamma(A, B)$ to compute $\gamma(G, H)$.

Feature 3: Lifting

$\mathrm{Hom}(G, H) \cong \mathrm{Hom}(A, B)$.

The framework can use $\gamma(A, B)$ to compute $\gamma(G, H)$.

G	H	Base Case
Any finite G	$\mathbb{Z}_p$	$\mathbb{Z}_p^n \to \mathbb{Z}_p$
Any finite dimensional Lie algebra over $\mathbb{F}_p$		
$\mathsf{GL}_n(\mathbb{F}_q)$	$\mathbb{F}_q^\times$	$\mathbb{Z}_{q-1} \to \mathbb{Z}_{q-1}$
$\mathfrak{gl}_n(\mathbb{F}_q)$	$\mathbb{F}_q$	$\mathbb{Z}_q \to \mathbb{Z}_q$

$\varphi : \mathbb{F}_q^{n \times n} \to \mathbb{F}_q, \quad \varphi(AB) = \varphi(BA)$

Feature 3: Lifting

$\mathrm{Hom}(G, H) \cong \mathrm{Hom}(A, B)$.

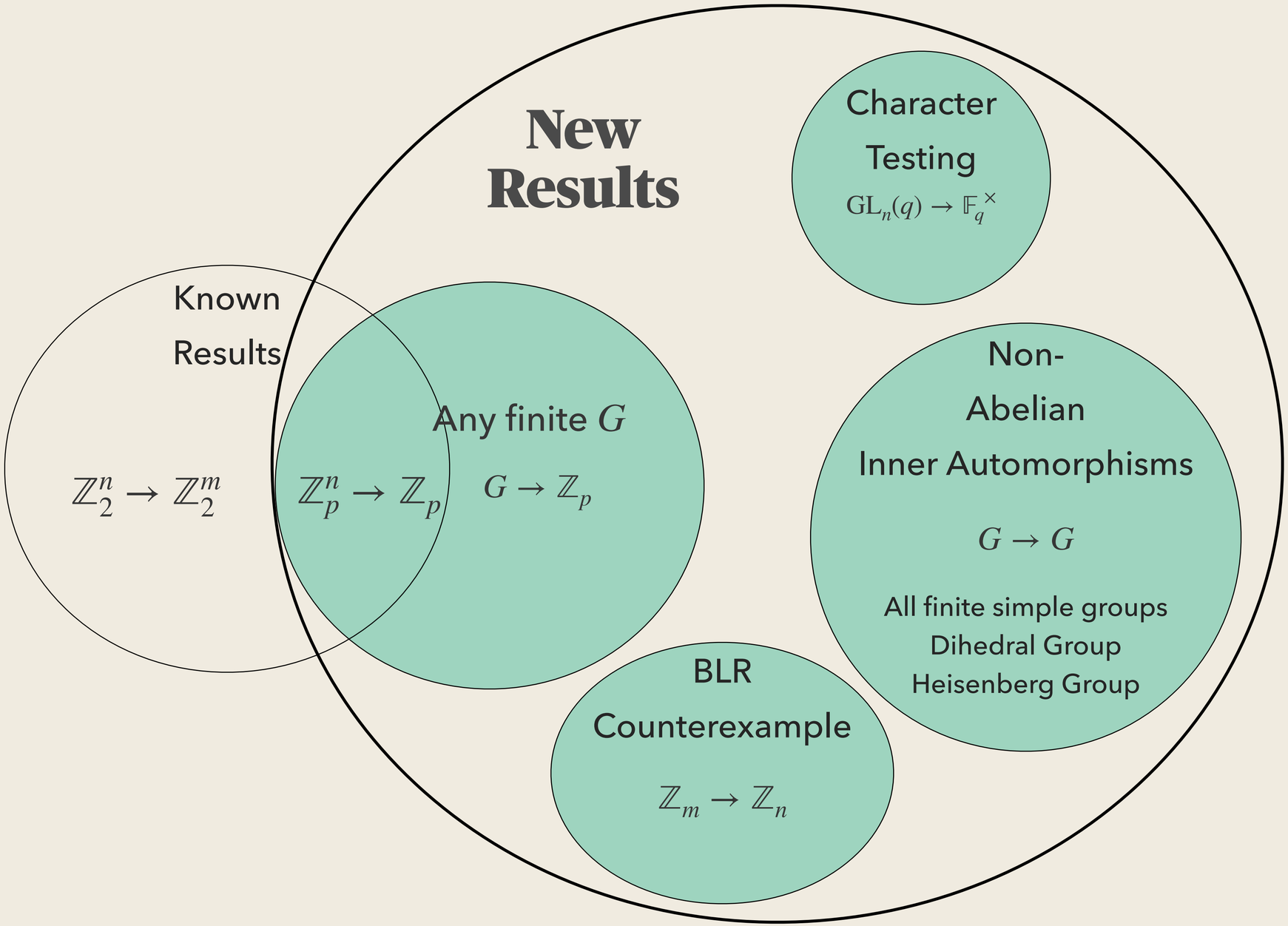
The framework can use $\gamma(A, B)$ to compute $\gamma(G, H)$.

Character Testing

G	H	Base Case
Any finite G	$\mathbb{Z}_p$	$\mathbb{Z}_p^n \to \mathbb{Z}_p$
Any finite dimensional Lie algebra over $\mathbb{F}_p$		
$\mathsf{GL}_n(\mathbb{F}_q)$	$\mathbb{F}_q^\times$	$\mathbb{Z}_{q-1} \to \mathbb{Z}_{q-1}$
$\mathfrak{gl}_n(\mathbb{F}_q)$	$\mathbb{F}_q$	$\mathbb{Z}_q \to \mathbb{Z}_q$

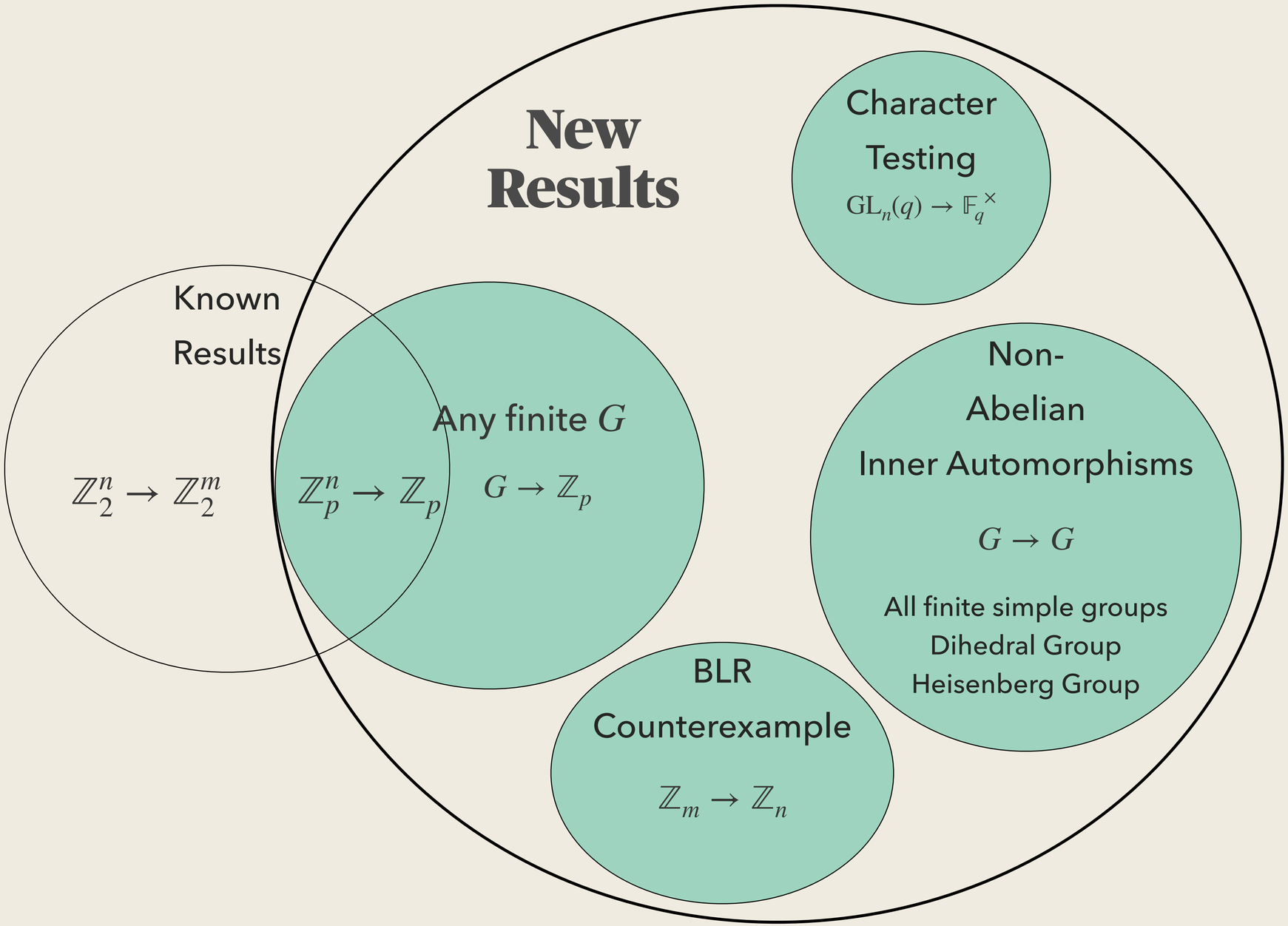
$\varphi : \mathbb{F}_q^{n \times n} \to \mathbb{F}_q, \quad \varphi(AB) = \varphi(BA)$

A General Framework



A General Framework

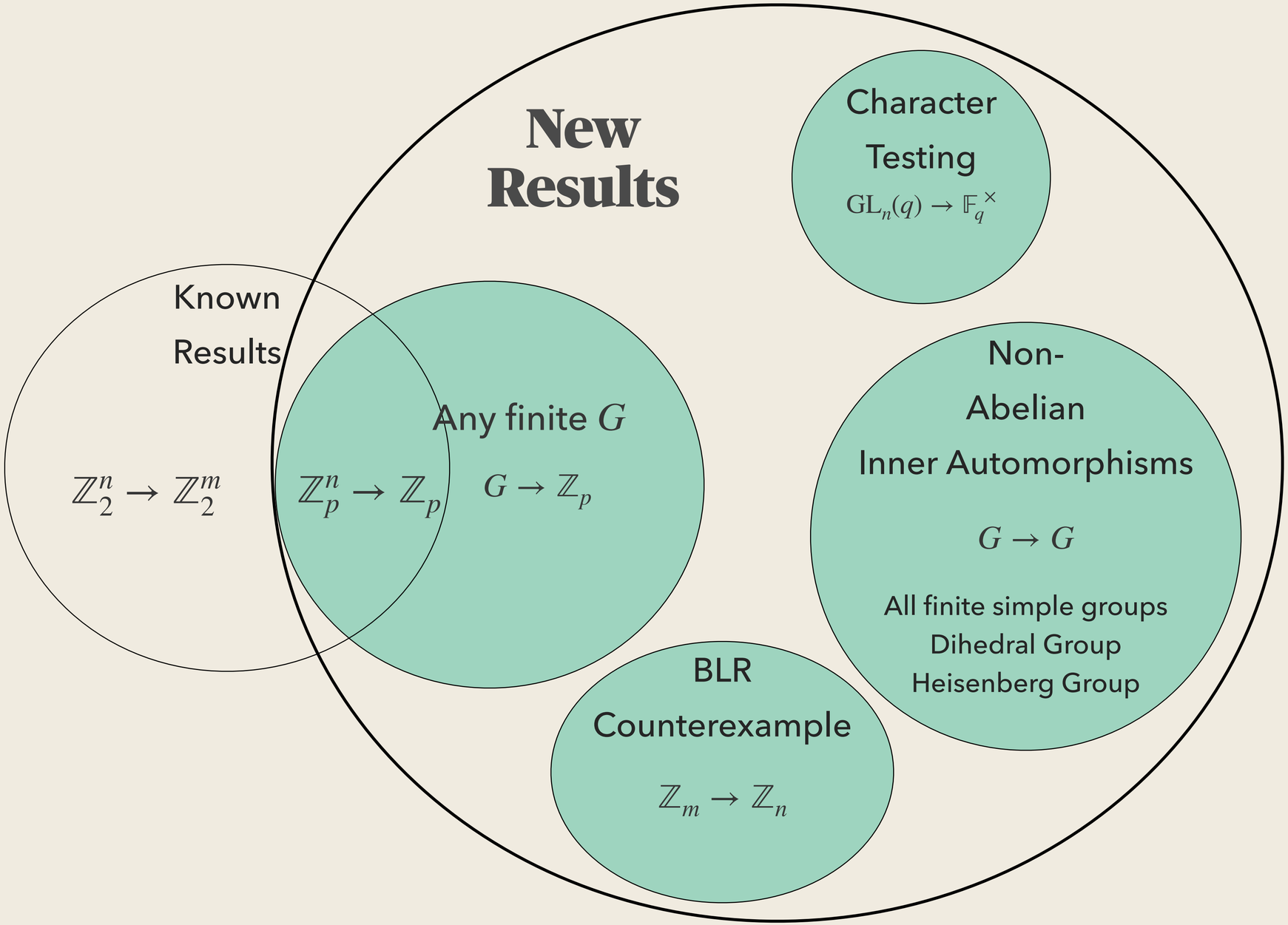
General but Group-Specific Tests



A General Framework

General but Group-Specific Tests

Flexible Query vs Soundness

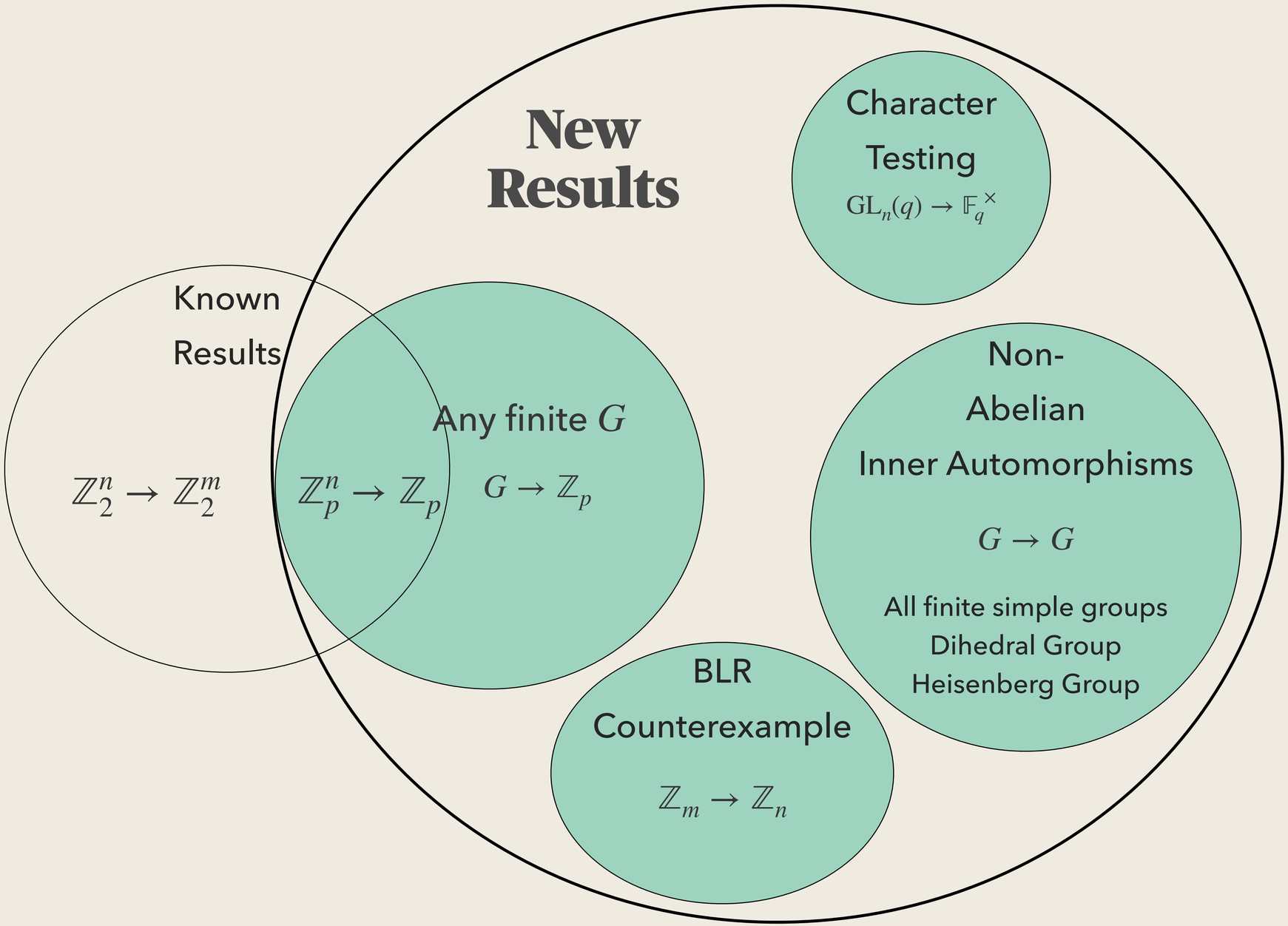


A General Framework

General but Group-Specific Tests

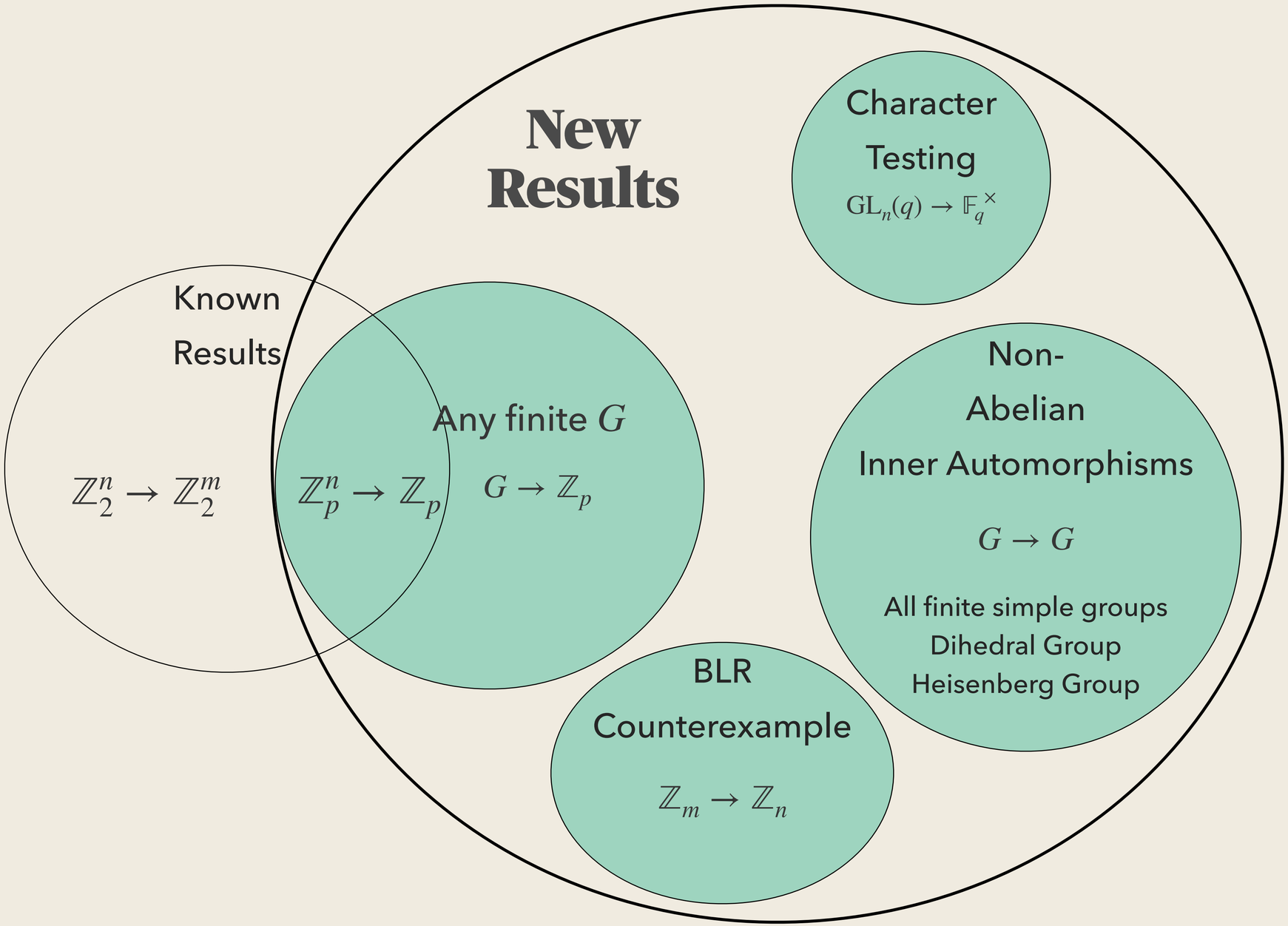
Flexible Query vs Soundness

List Decoding Bounds



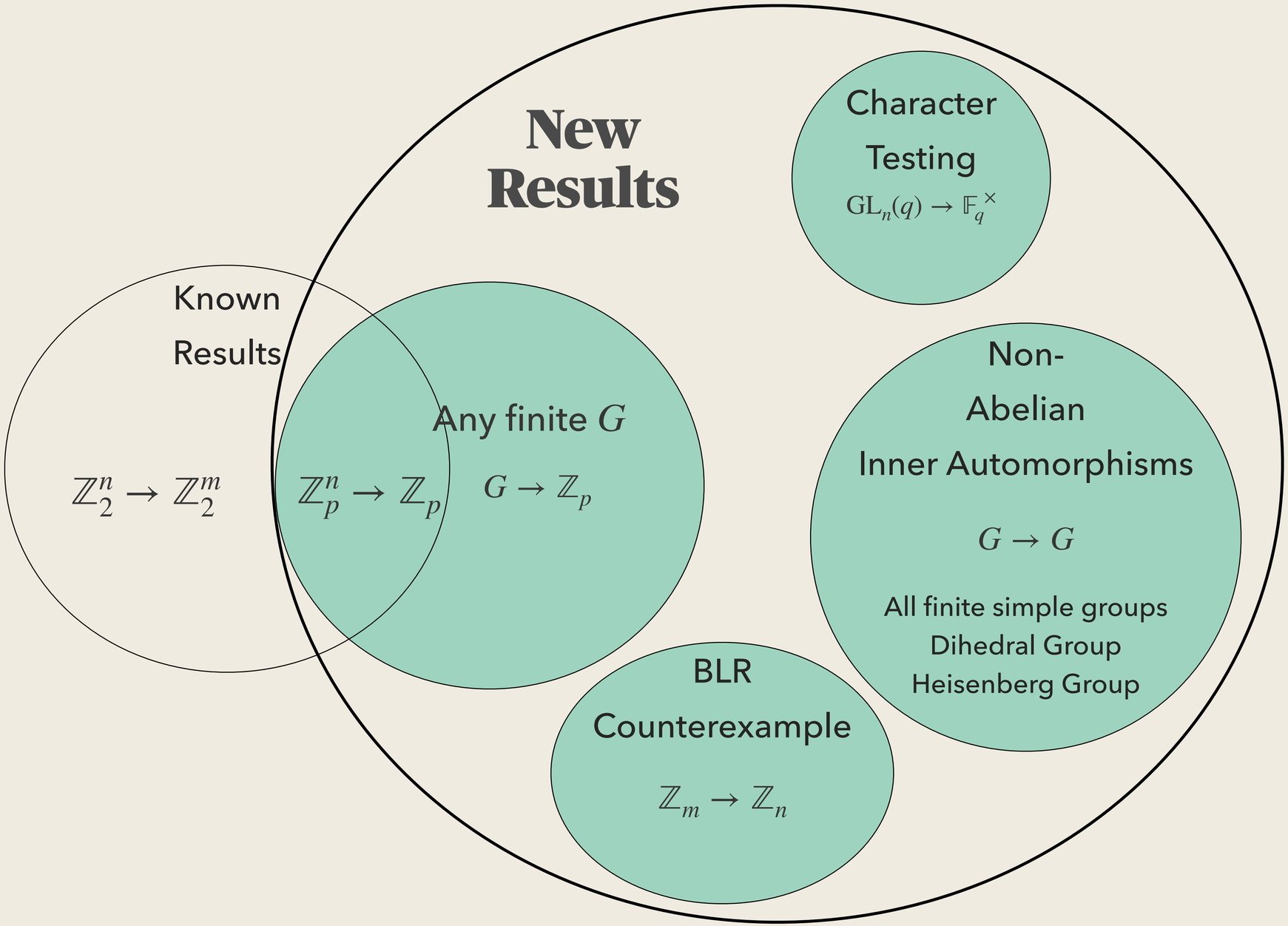
A General Framework

- General but Group-Specific Tests
- Flexible Query vs Soundness
- List Decoding Bounds
- “Lifting” Property



A General Framework

- General but Group-Specific Tests
- Flexible Query vs Soundness
- List Decoding Bounds
- “Lifting” Property
- Fourier-free Proof



Open Problems

- Investigating more groups.
- Reducing random bits needed—*derandomizing* via HDXs?
- Tackling other (structured) function classes.

Thank You!